Experiments in Astroparticle Physics
(Space-based experiments)
(Lecture 11)

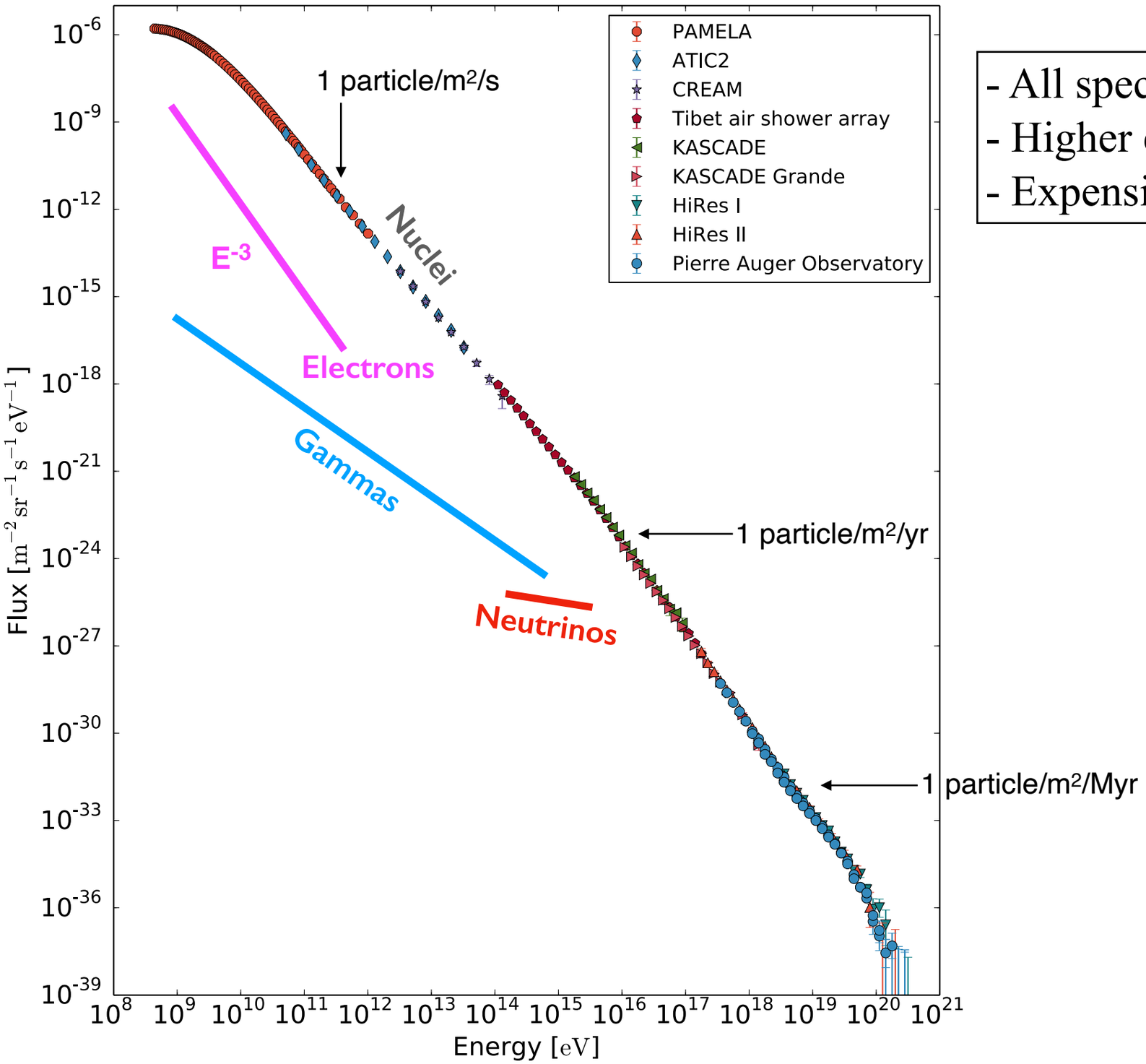
Satyendra Thoudam
Khalifa University

Outline of this chapter

- Introduction
- Measurement techniques:
    - Space-based
    - Ground-based
- Space-based experiments:
    - AMS-02 low-energy cosmic-ray experiment
    - FERMI gamma-ray telescope
    - JEM-EUSO ultra-high-energy cosmic-ray detector

Introduction: Where do we build the detectors?

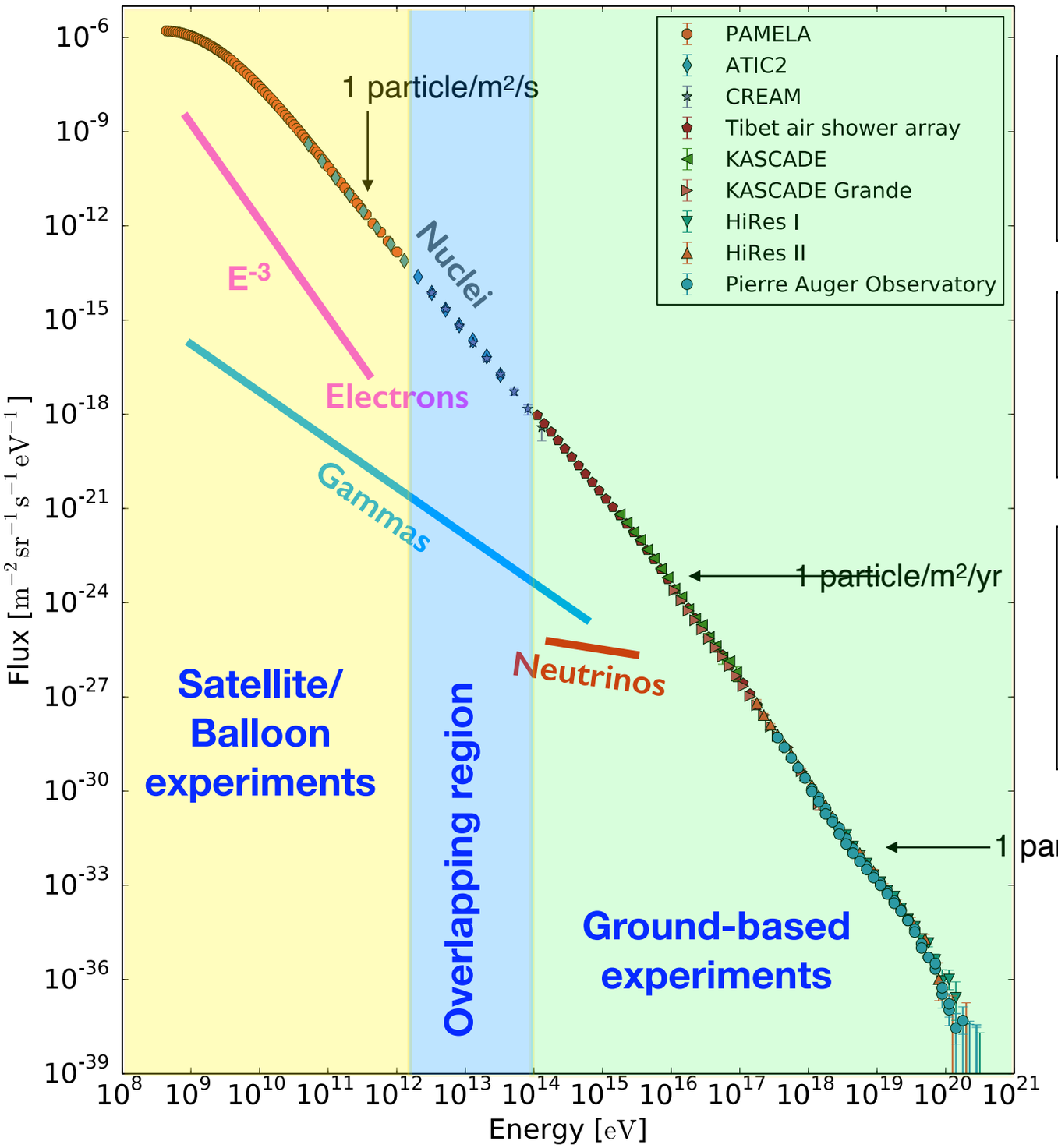
Spectra of cosmic rays, gamma rays and neutrinos



- All spectra follow a steep power law.
- Higher energy —-> Bigger detectors.
- Expensive to send bigger detector in space.

Introduction: Where do we build the detectors?

Spectra of cosmic rays, gamma rays and neutrinos



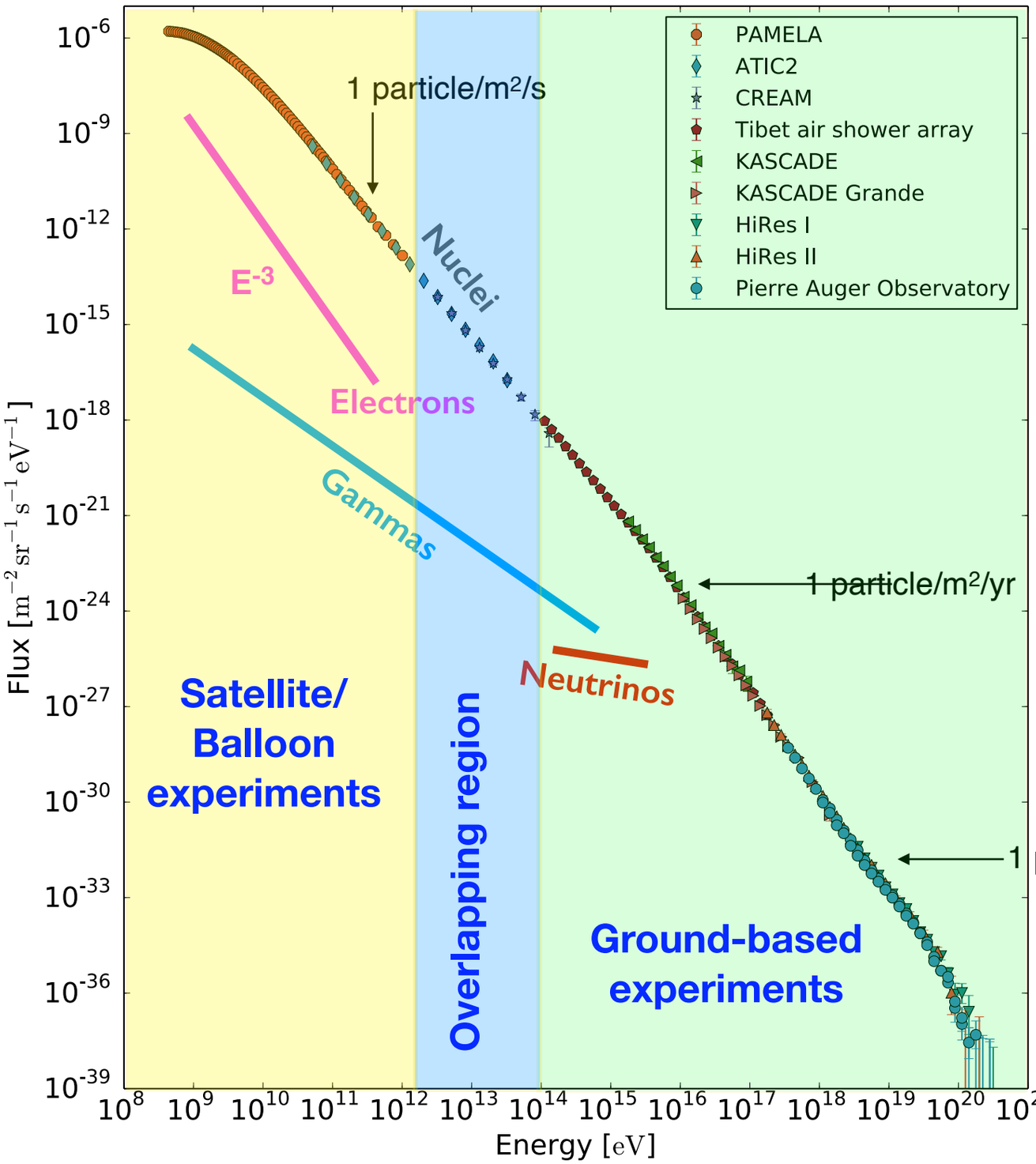
- All spectra follow a steep power law.
- Higher energy —-> Bigger detectors.
- Expensive to send bigger detector in space.

- For CR nuclei, up to ~ 10^{14} eV is mainly space-based experiment. Above is mostly ground-based.

- For electrons and gammas, only up to ~ 10^{12} eV is mainly space-based experiment.
- This is less than nuclei because of their smaller fluxes.

Introduction: Where do we build the detectors?

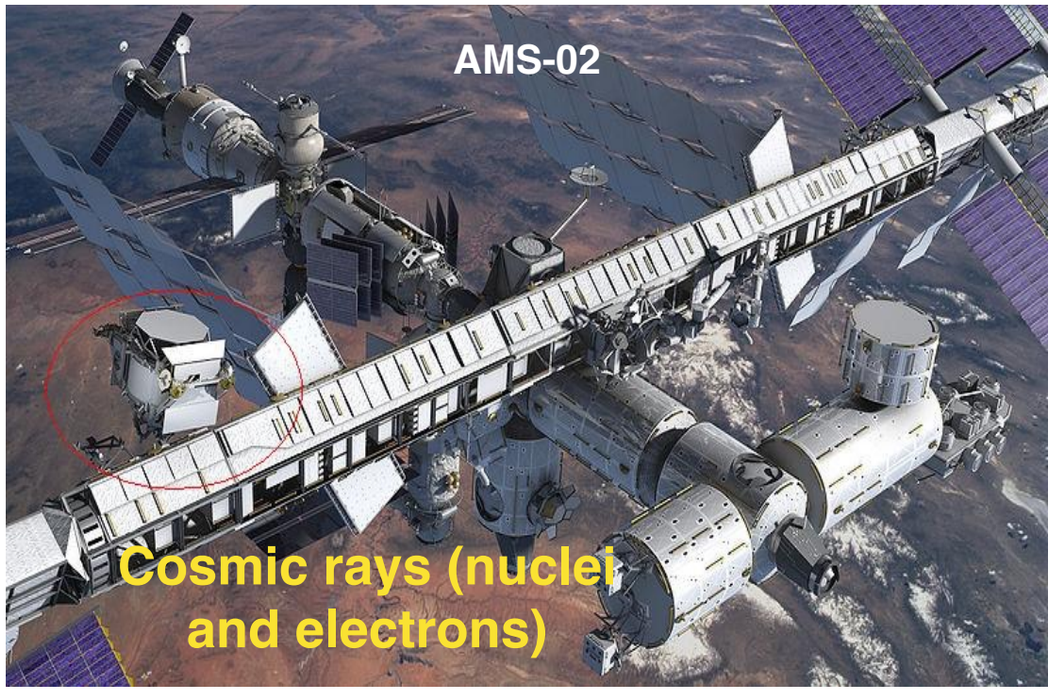
Spectra of cosmic rays, gamma rays and neutrinos



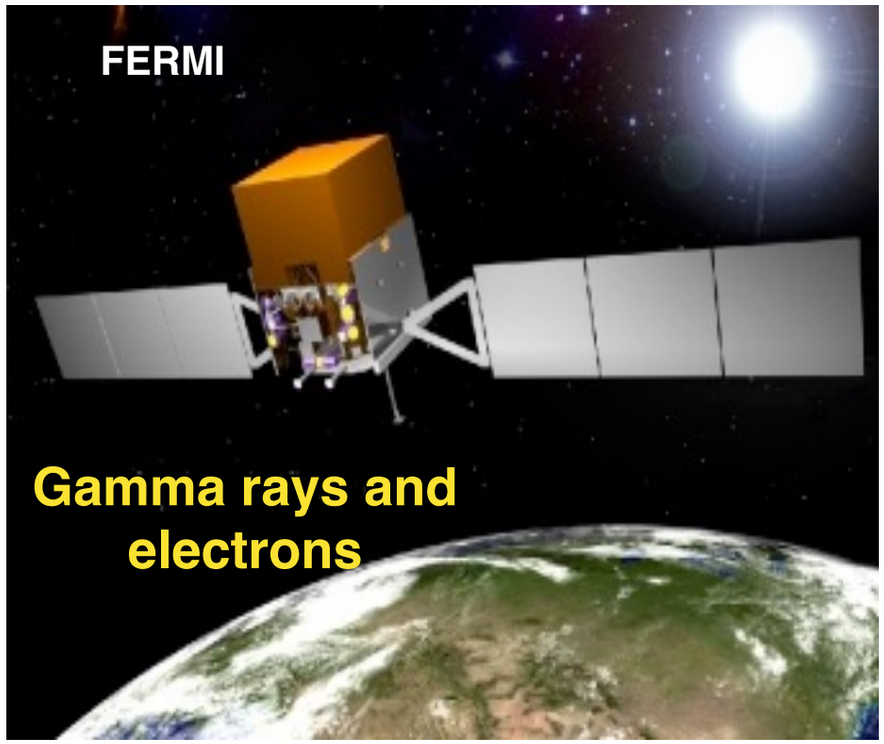
*Basic aim of every experiment is to measure the **Energy, Charge (or type) and Direction (or momentum)** of the incoming particle.

Measurements techniques: Spaced-based experiments

Satellite experiments

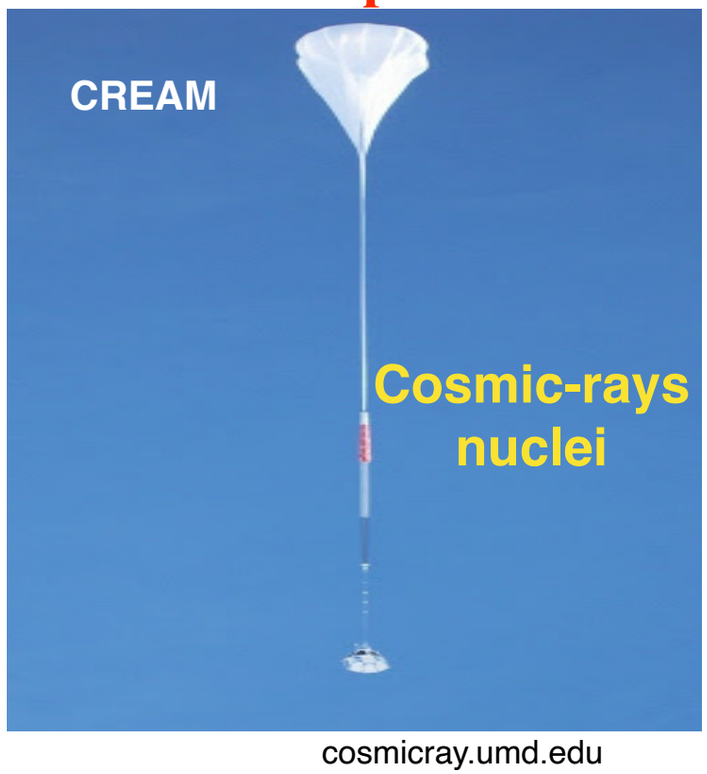


www.ams02.org

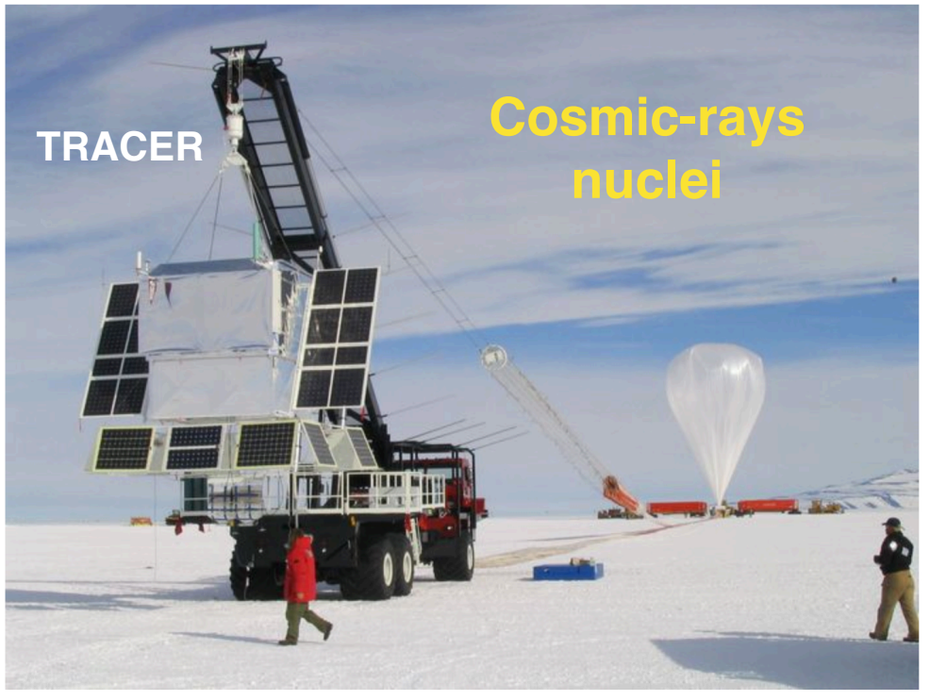


fermi.gsfc.nasa.gov

Balloon experiments



cosmicray.umd.edu



University of Chicago

The basic principle used in satellite-based and balloon-based experiments are quite similar.

Measurements techniques: Spaced-based experiments

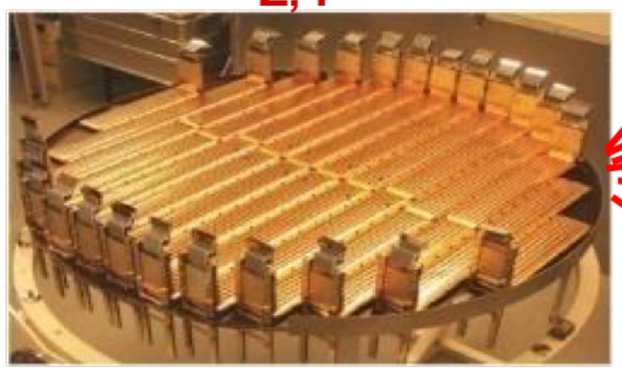
AMS-02 experiment on the International Space Station

(Most precise spaced-based CR experiment ever built!)

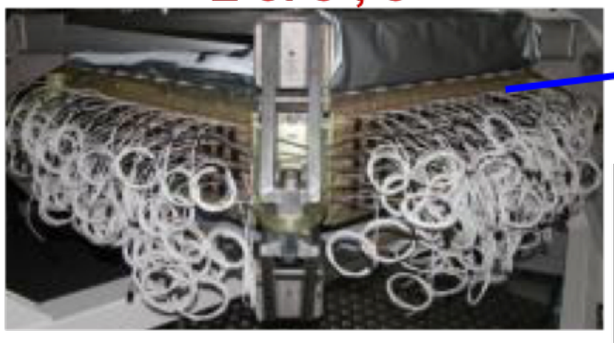
TRD
Identify e^+, e^-

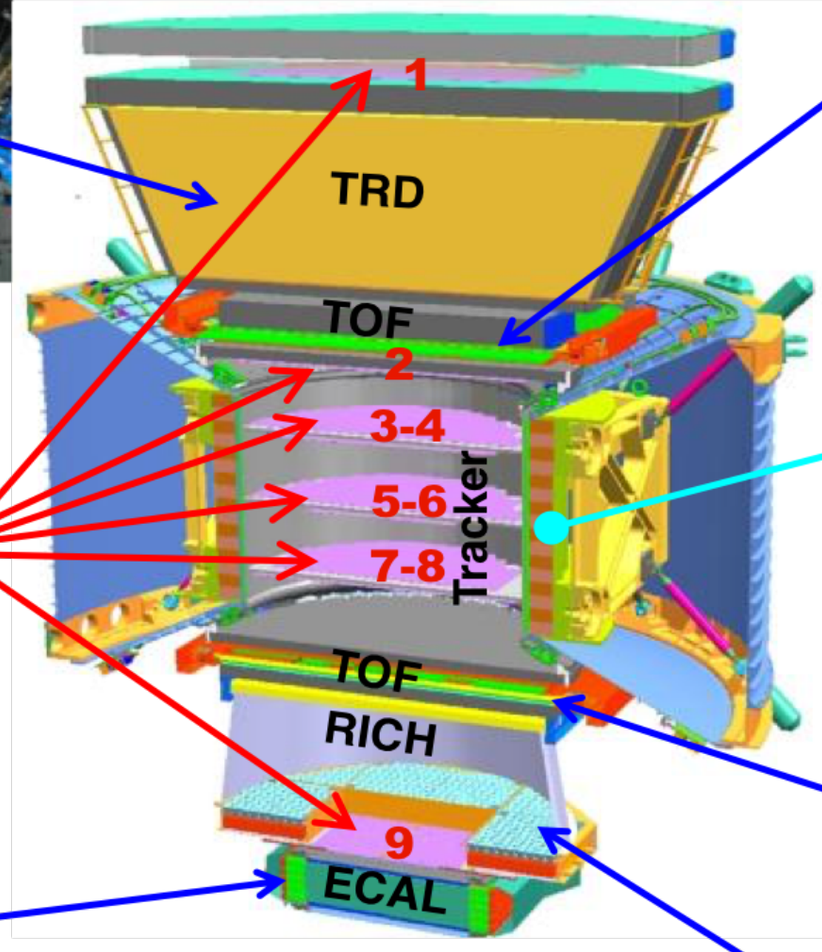


Silicon Tracker
Z, P



ECAL
E of e^+, e^-

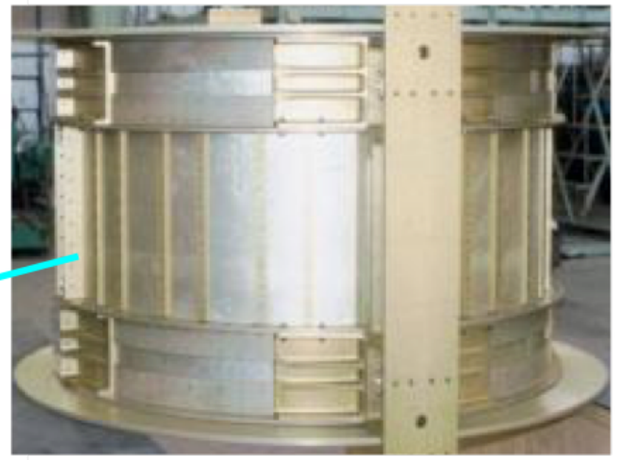




Time of Flight
Z, E



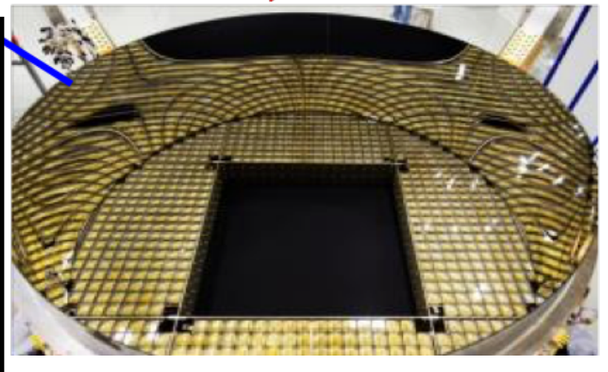
Magnet
±Z, P



Time of Flight
Z, E



Ring Imaging Cherenkov
Z, E



Tracker: Silicon Tracker.
TRD: Transition Radiation Detector.
ToF: Time of Flight counters.
MAGNET: A permanent magnet.
ACC: AntiCoincidence Counters.
RICH: Ring imaging Cherenkov detector.
ECAL: Electromagnetic Calorimeter.

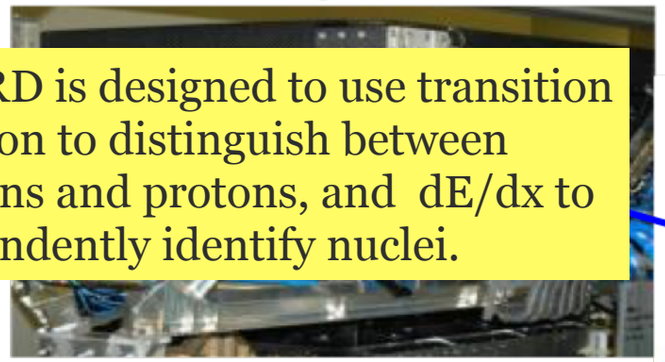
https://ams02.space/detector

Measurements techniques: Spaced-based experiments

AMS-02 experiment on the International Space Station

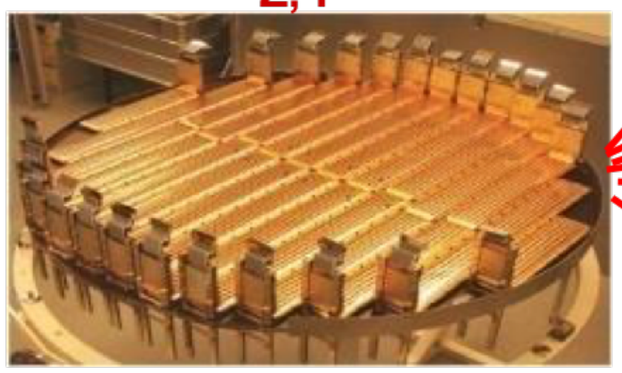
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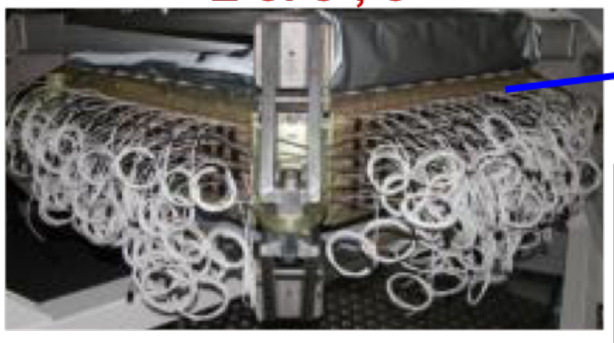


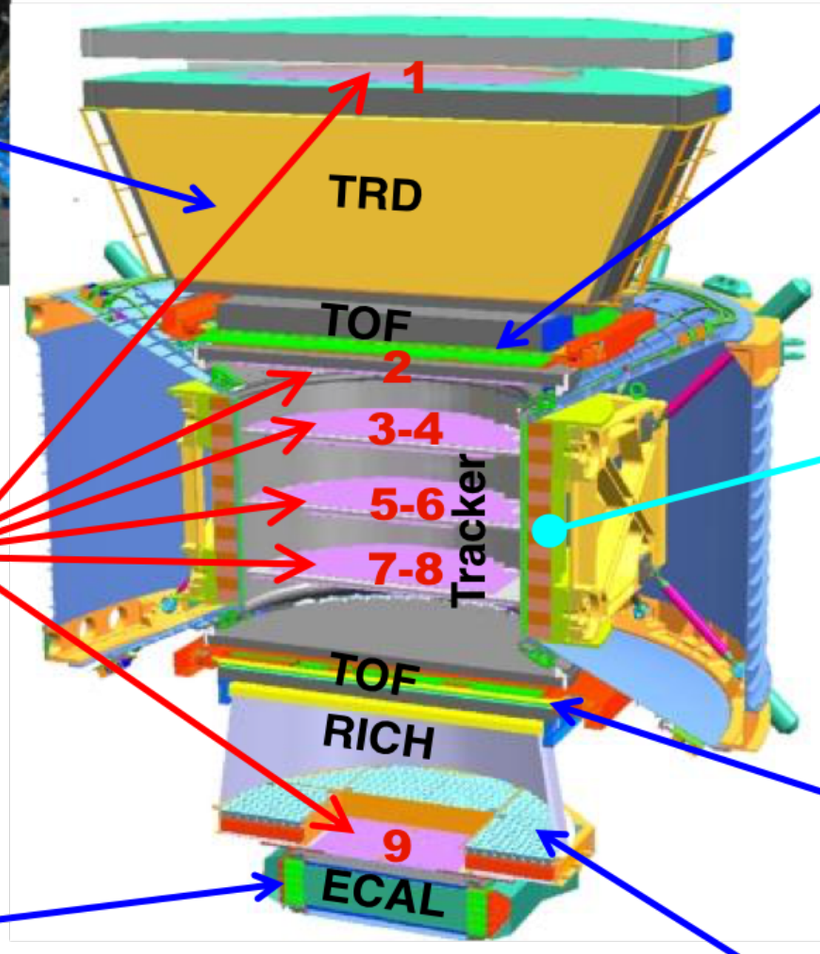
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Silicon Tracker
Z, P



ECAL
E of e^+, e^-





Time of Flight
Z, E



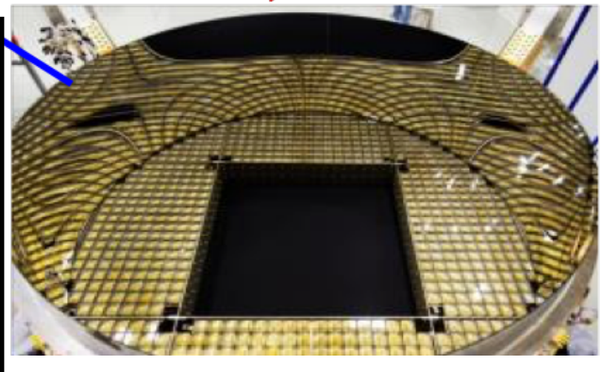
Magnet
±Z, P



Time of Flight
Z, E



Ring Imaging Cherenkov
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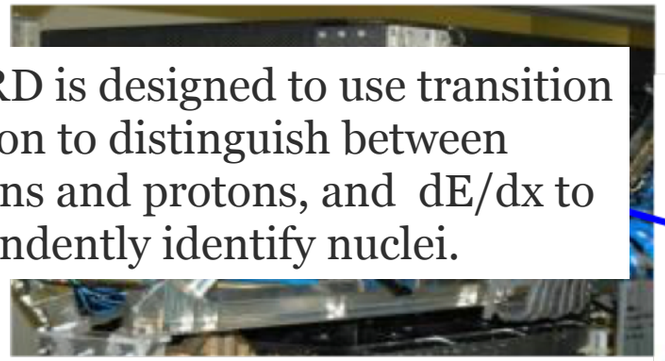
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Measurements techniques: Spaced-based experiments

AMS-02 experiment on the International Space Station

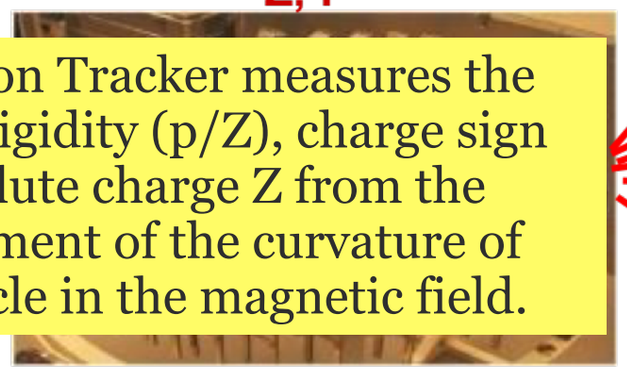
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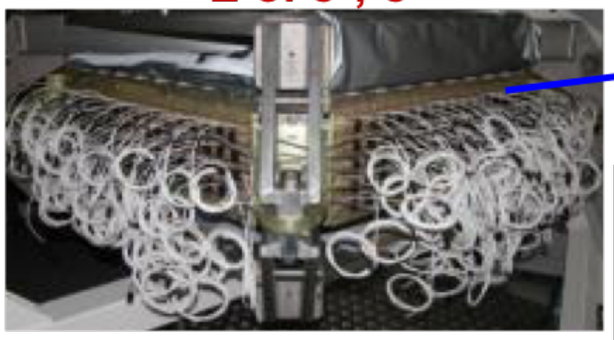
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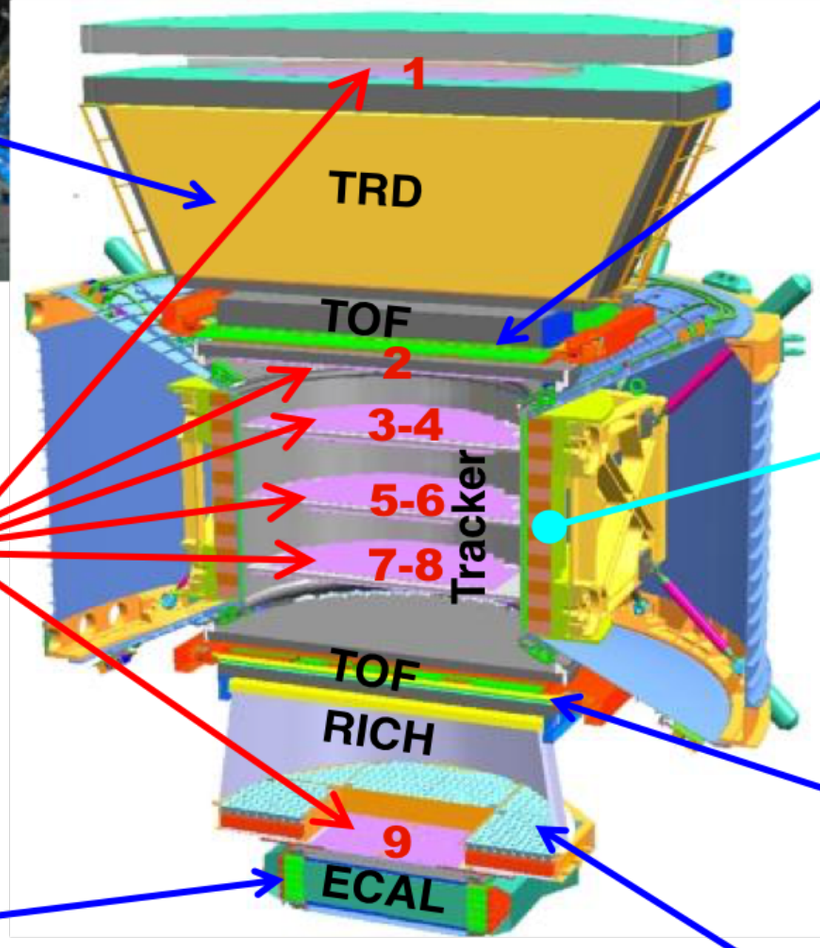
Silicon Tracker
Z, P



The Silicon Tracker measures the particle rigidity (p/Z), charge sign and absolute charge Z from the measurement of the curvature of the particle in the magnetic field.

ECAL
E of e^+, e^-

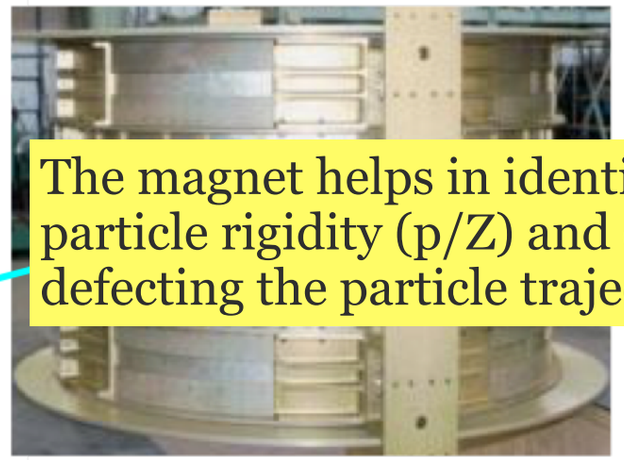




Time of Flight
Z, E



Magnet
±Z, P

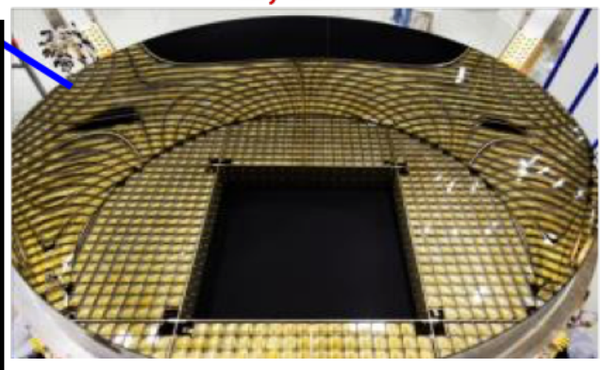


The magnet helps in identifying the particle rigidity (p/Z) and charge by defecting the particle trajectory.

Time of Flight
Z, E



Ring Imaging Cherenkov
Z, E



Tracker: Silicon Tracker.
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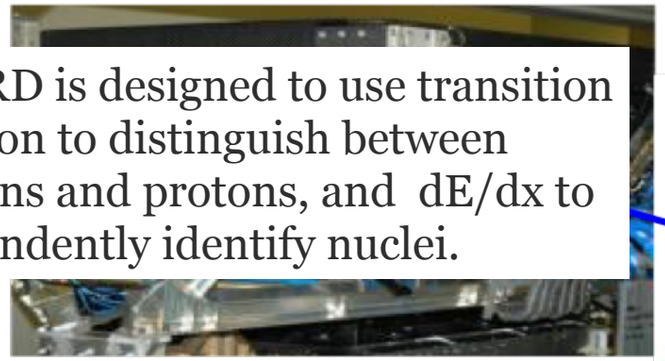
https://ams02.space/detector

Measurements techniques: Spaced-based experiments

AMS-02 experiment on the International Space Station

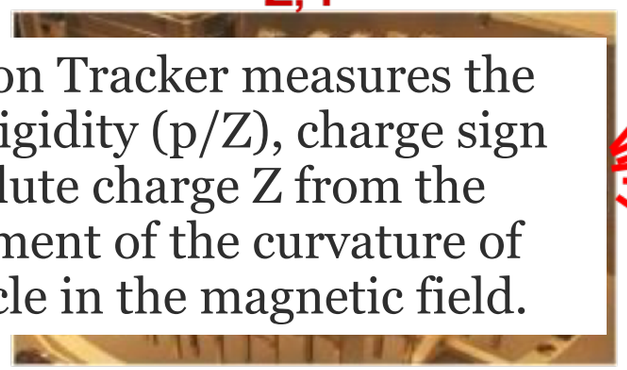
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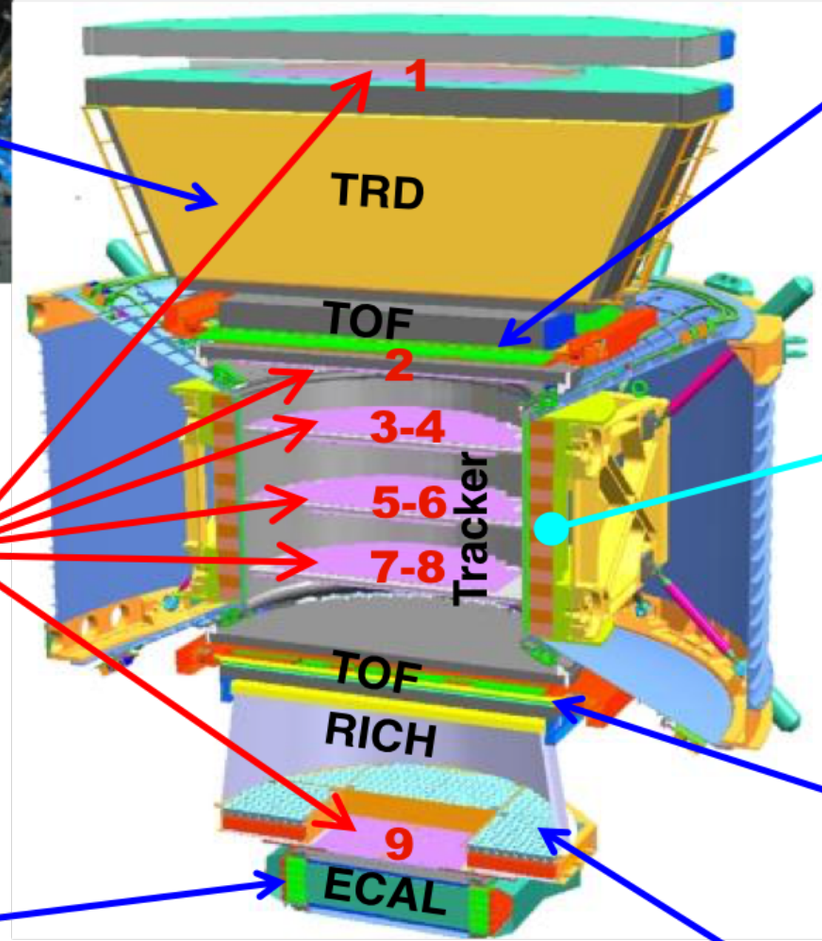
Silicon Tracker
Z, P



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ECAL
E of e^+, e^-





Time of Flight
Z, E

Two planes of TOF counters are used for trigger and also to discriminate between downward and upward-going particles.

Magnet
±Z, P



The magnet helps in identifying the particle rigidity (p/Z) and charge by defecting the particle trajectory.

Time of Flight
Z, E



Ring Imaging Cherenkov
Z, E



The RICH measures the charge and velocity.

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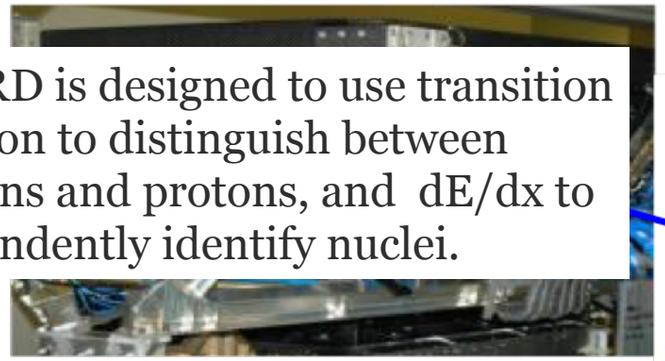
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Measurements techniques: Spaced-based experiments

AMS-02 experiment on the International Space Station

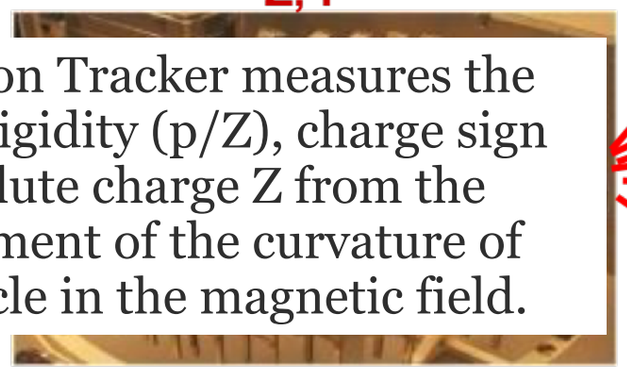
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TRD
Identify e^+, e^-



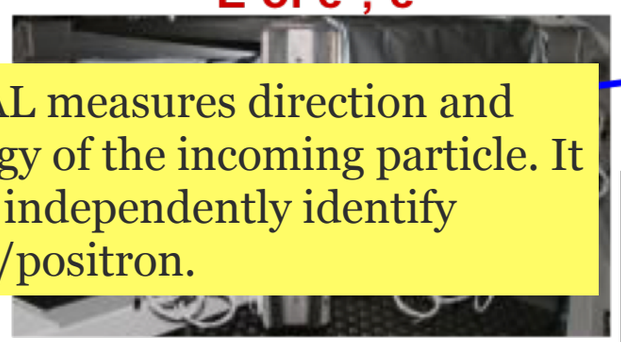
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Silicon Tracker
Z, P

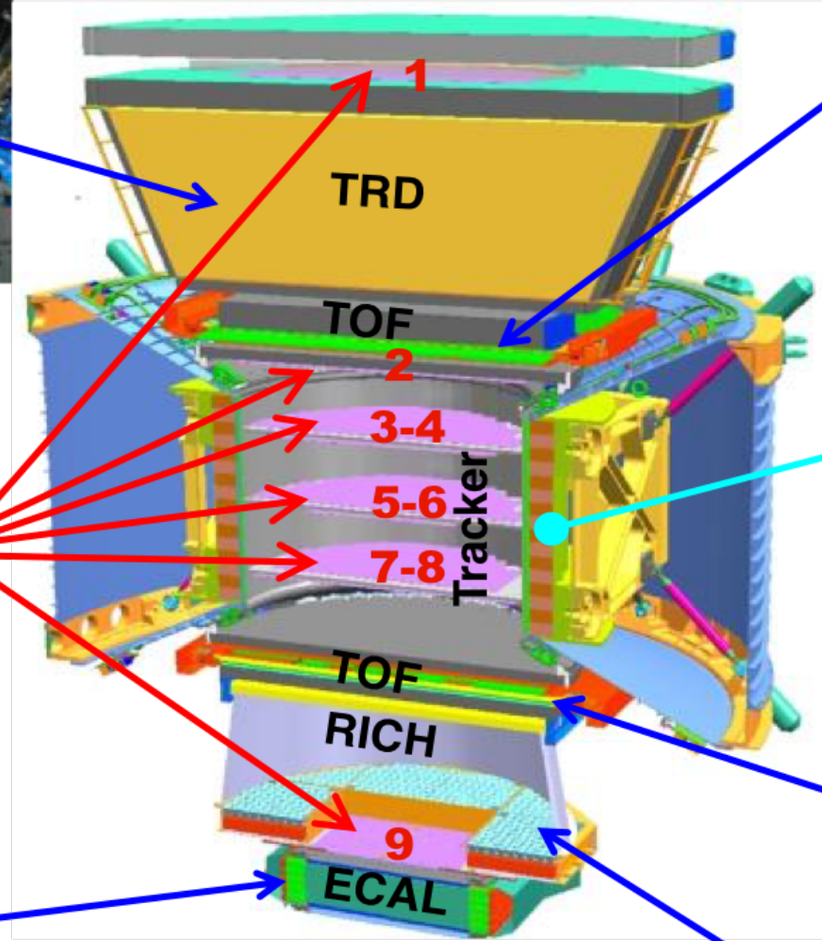


The Silicon Tracker measures the particle rigidity (p/Z), charge sign and absolute charge Z from the measurement of the curvature of the particle in the magnetic field.

ECAL
E of e^+, e^-



The ECAL measures direction and the energy of the incoming particle. It can also independently identify electron/positron.



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Time of Flight
Z. E

Two planes of TOF counters are used for trigger and also to discriminate between downward and upward-going particles.

Magnet
±Z, P



The magnet helps in identifying the particle rigidity (p/Z) and charge by defecting the particle trajectory.



Time of Flight
Z, E



Ring Imaging Cherenkov
Z, E



The RICH measures the charge and velocity.

https://ams02.space/detector

Measurements techniques: Spaced-based experiments

* **AMS-02 has collected over 200 billion cosmic-ray events since its launch in 2011.**
* **Shown is a measured 369 GeV positron event.**

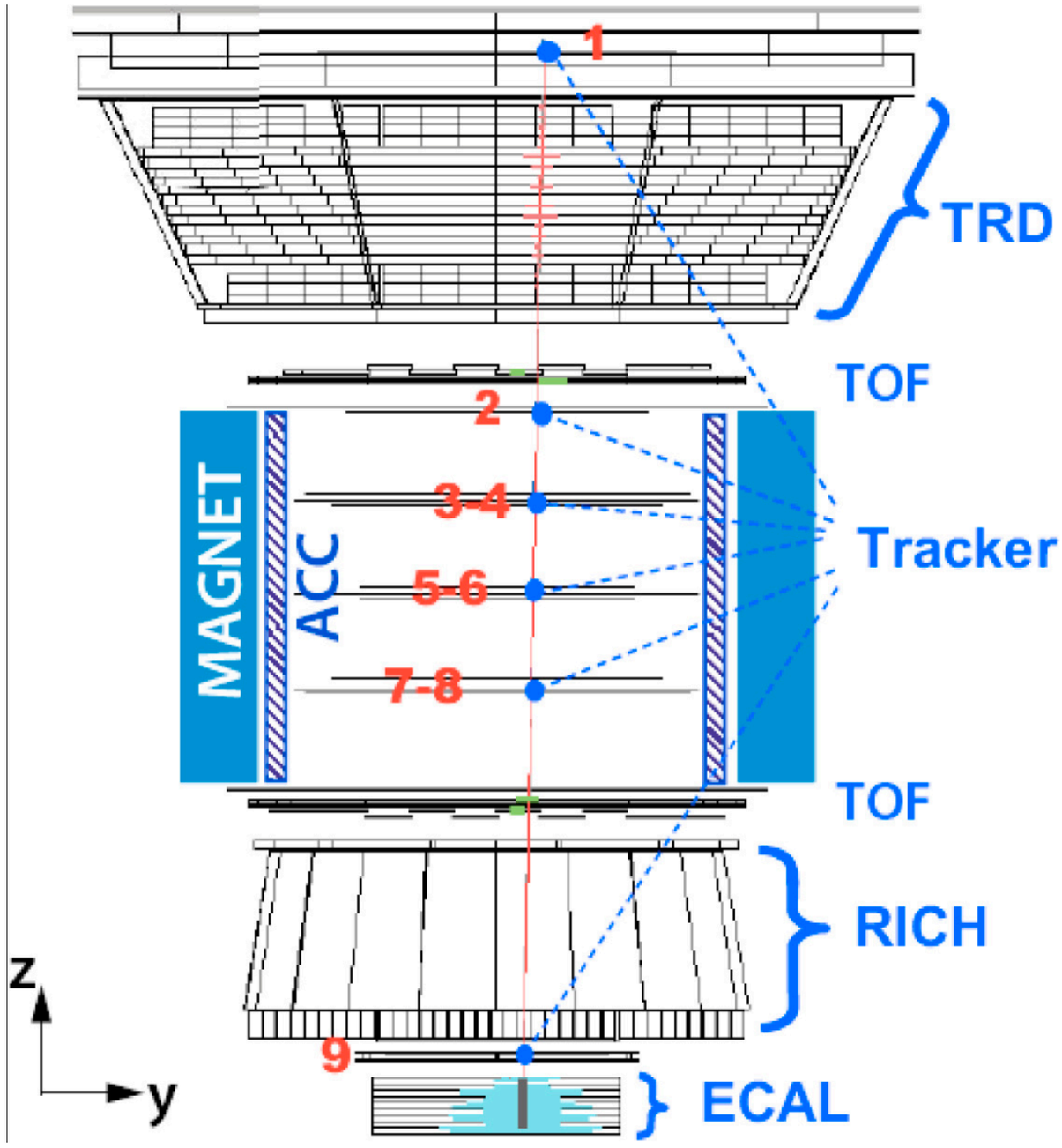
- Tracker planes 1–9 measure the particle **charge and momentum.**

- The TRD identifies the particle as an **electron/positron.**

- The ToF measures the charge and ensures that the particle is **downward-going**.

- The RICH measures the **charge and velocity**.

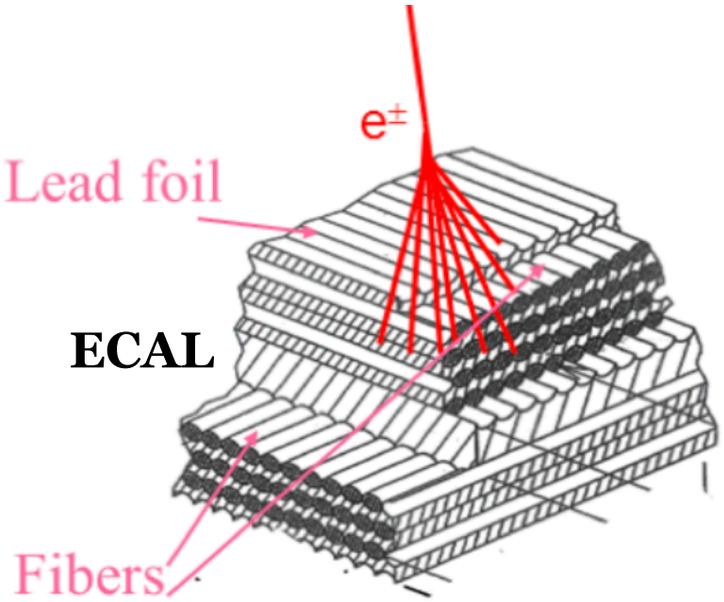
- The ECAL independently identifies the particles an **electron/positron** and measures its **energy**.

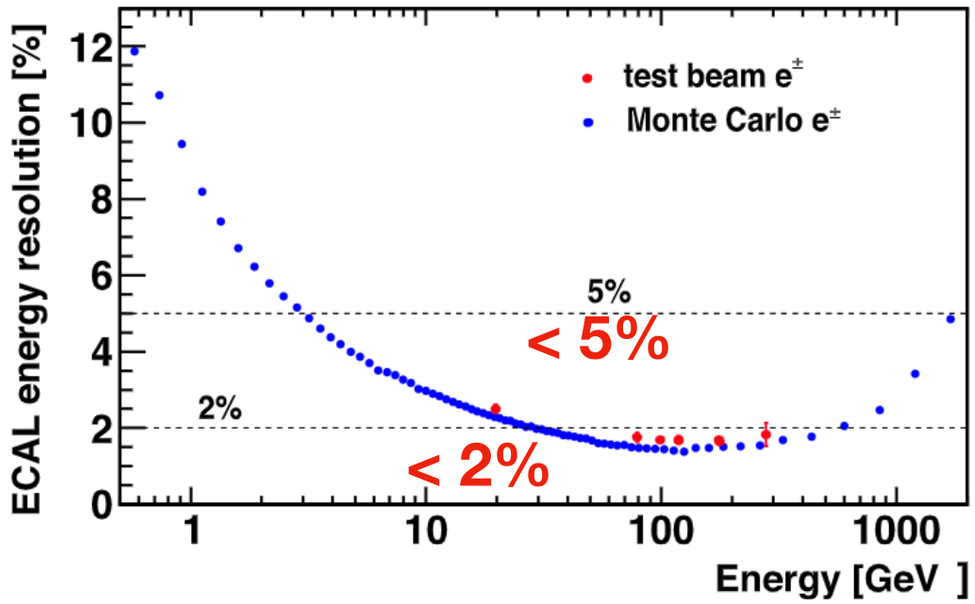


Measurements techniques: Spaced-based experiments

Performance of AMS-02

Energy resolution

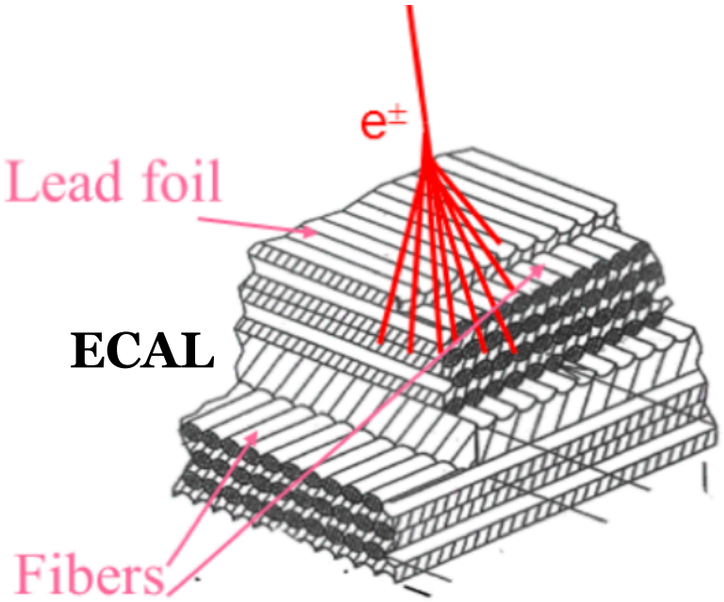


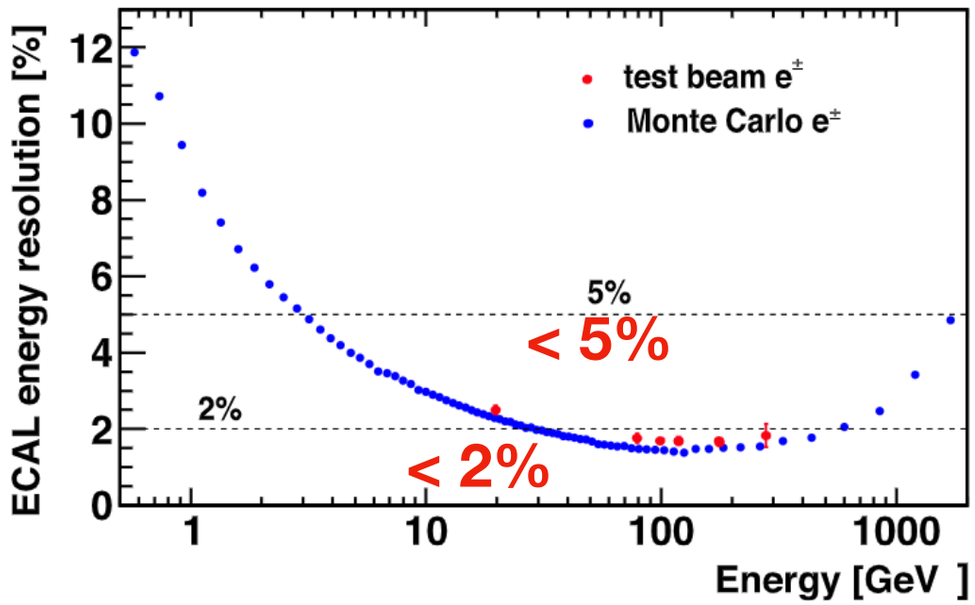


Measurements techniques: Spaced-based experiments

Performance of AMS-02

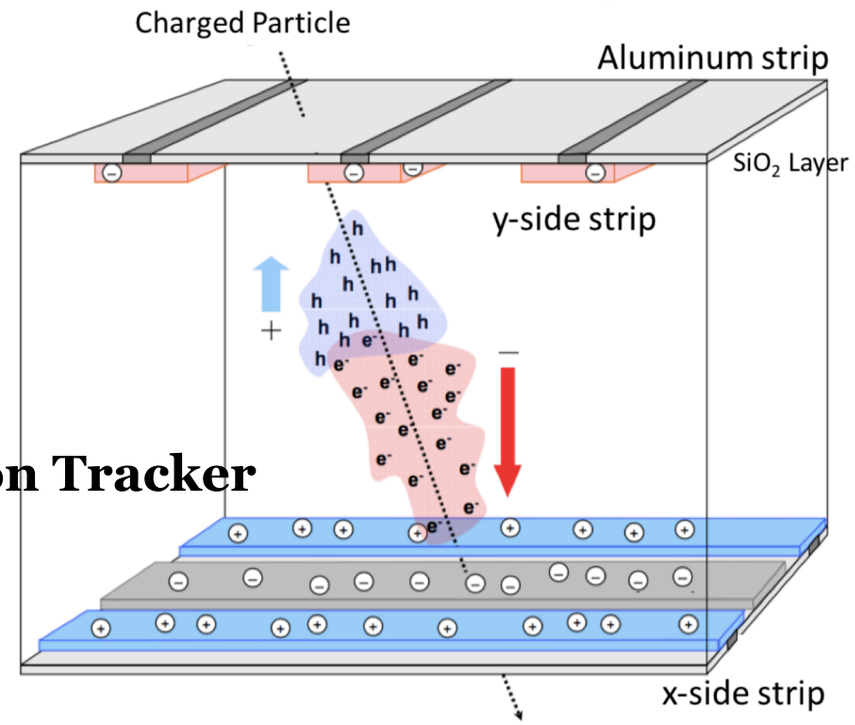
Energy resolution

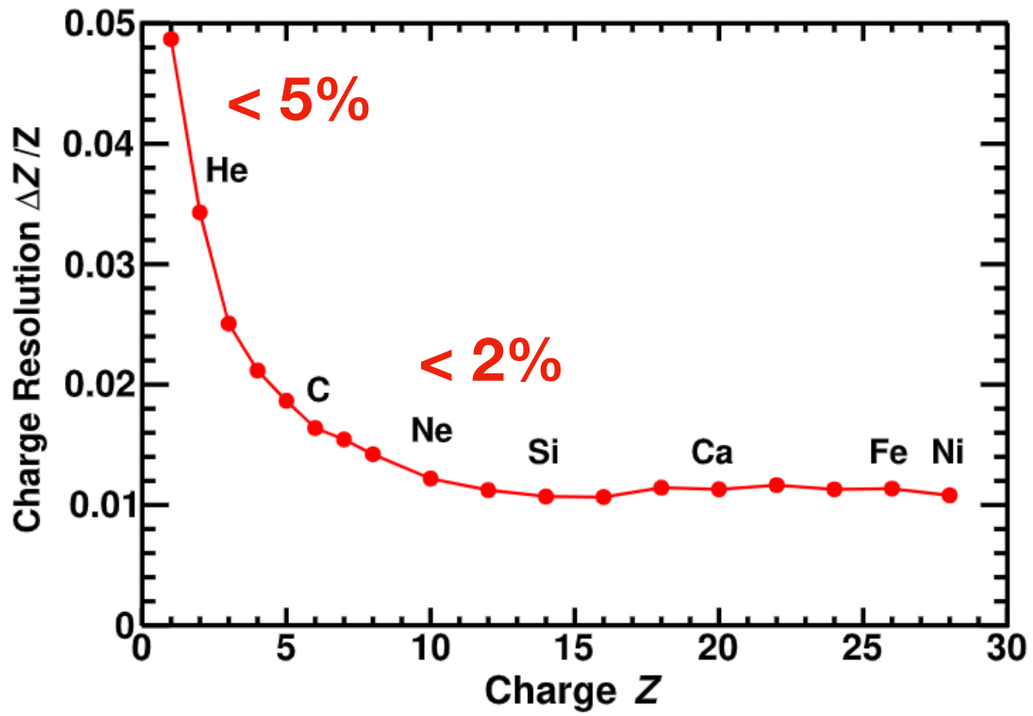




Charge resolution

Silicon Tracker





AMS-02 Collaboration, 2021, Physics Reports, 1

Measurements techniques: Spaced-based experiments

Performance of AMS-02

Systematic error:
- Uncertainties in the trigger efficiency: < 1.5%
- Event acceptance: < 5%
- Geomagnetic cutoff factor: < 2%
- Unfolding: < 3%
- Rigidity resolution function and the absolute rigidity scale: < 5%
- …..

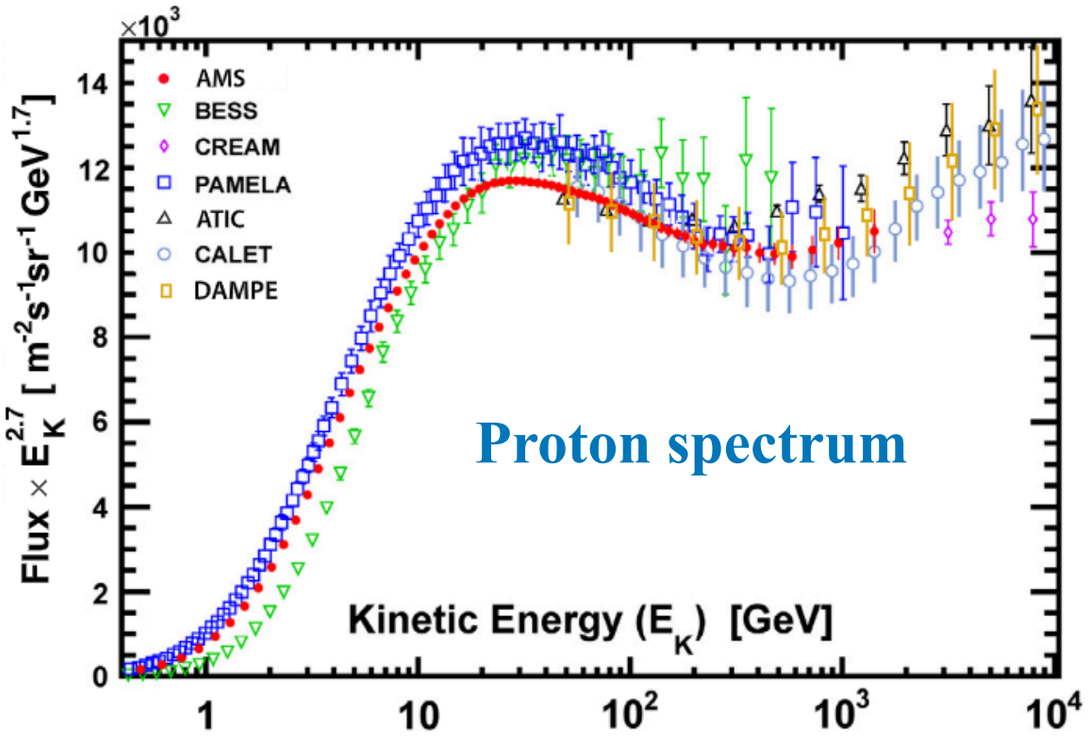
These result into a total systematic uncertainty on the measured cosmic-ray flux below 5%.

AMS-02 Collaboration, 2021, Physics Reports, 1

Measurements techniques: Spaced-based experiments

Some science results from AMS-02 measurements

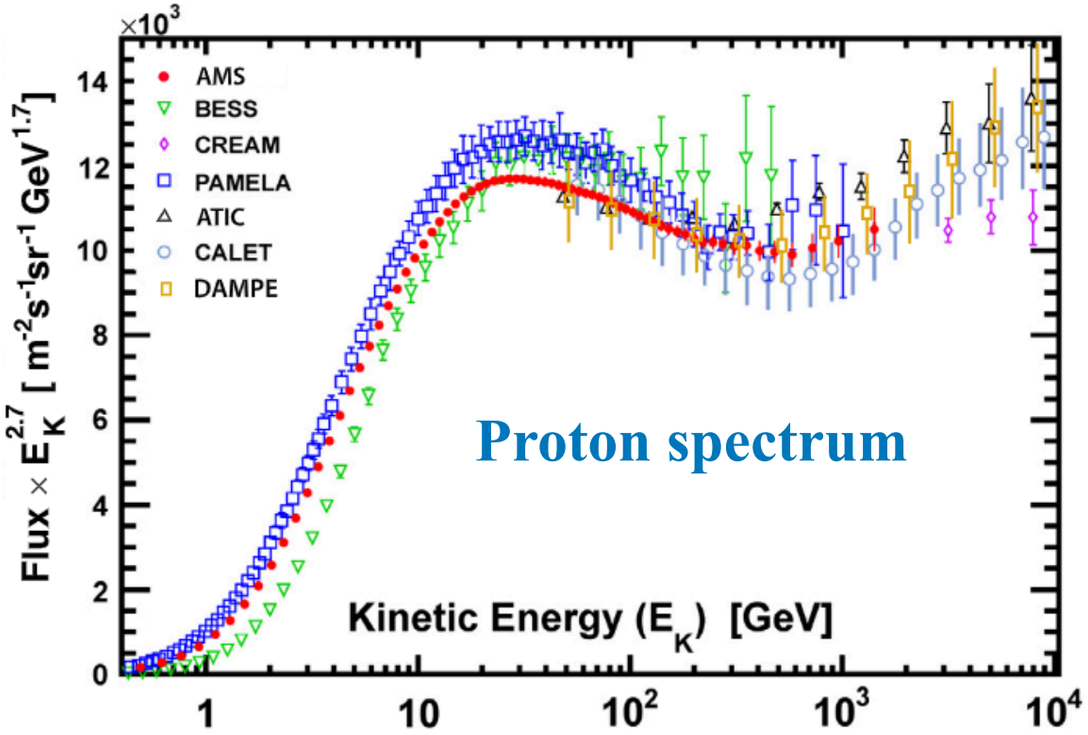
The highest precision CR measurements below 10^{13} eV till date!

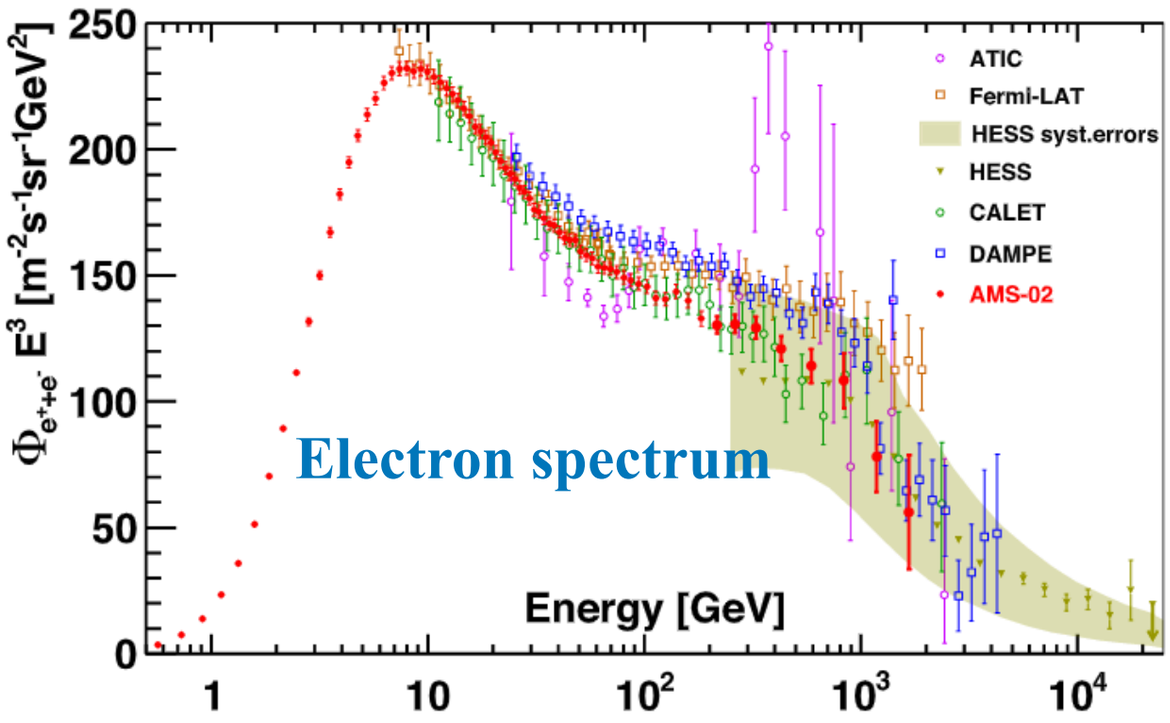


Measurements techniques: Spaced-based experiments

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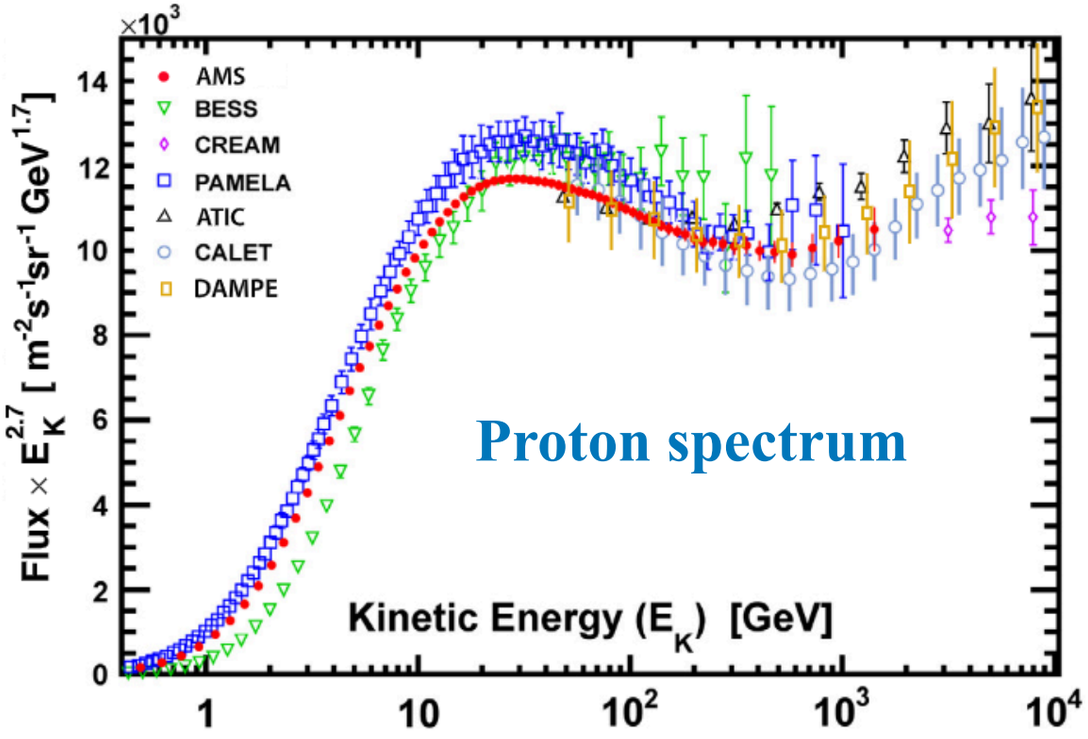


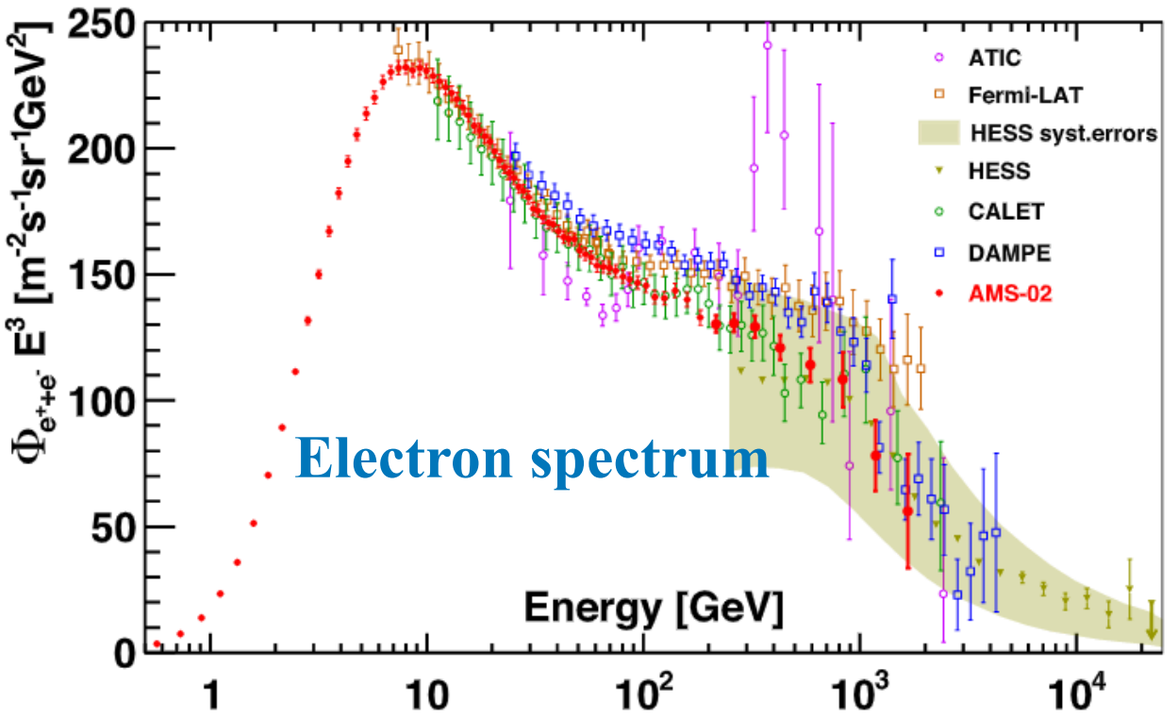
AMS-02 Collaboration, 2021, Physics Reports, 1

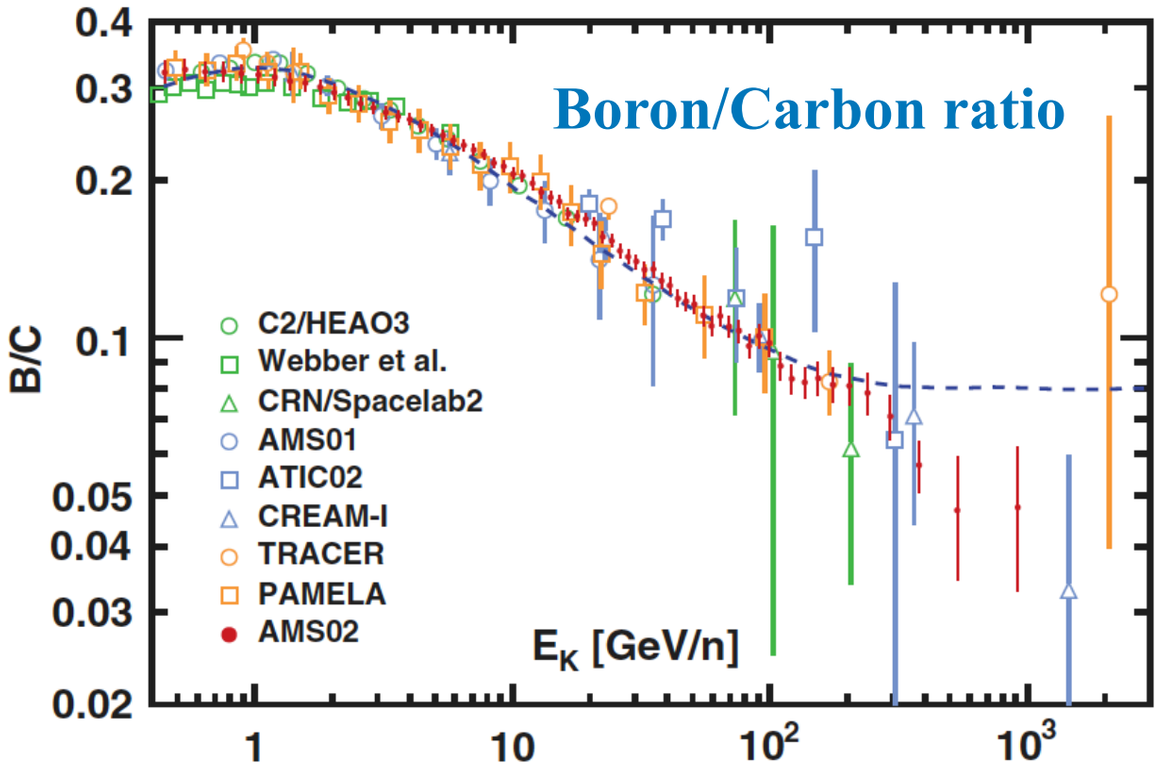
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Some science results from AMS-02 measurements

The highest precision CR measurements below 10^{13} eV till date!





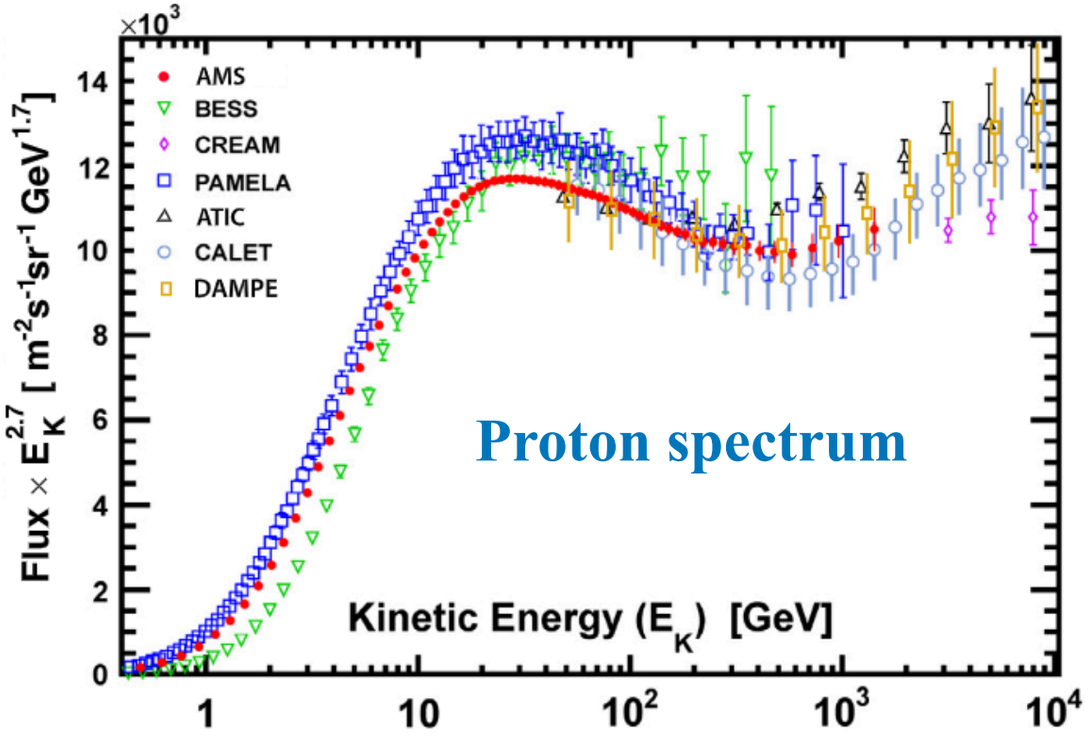


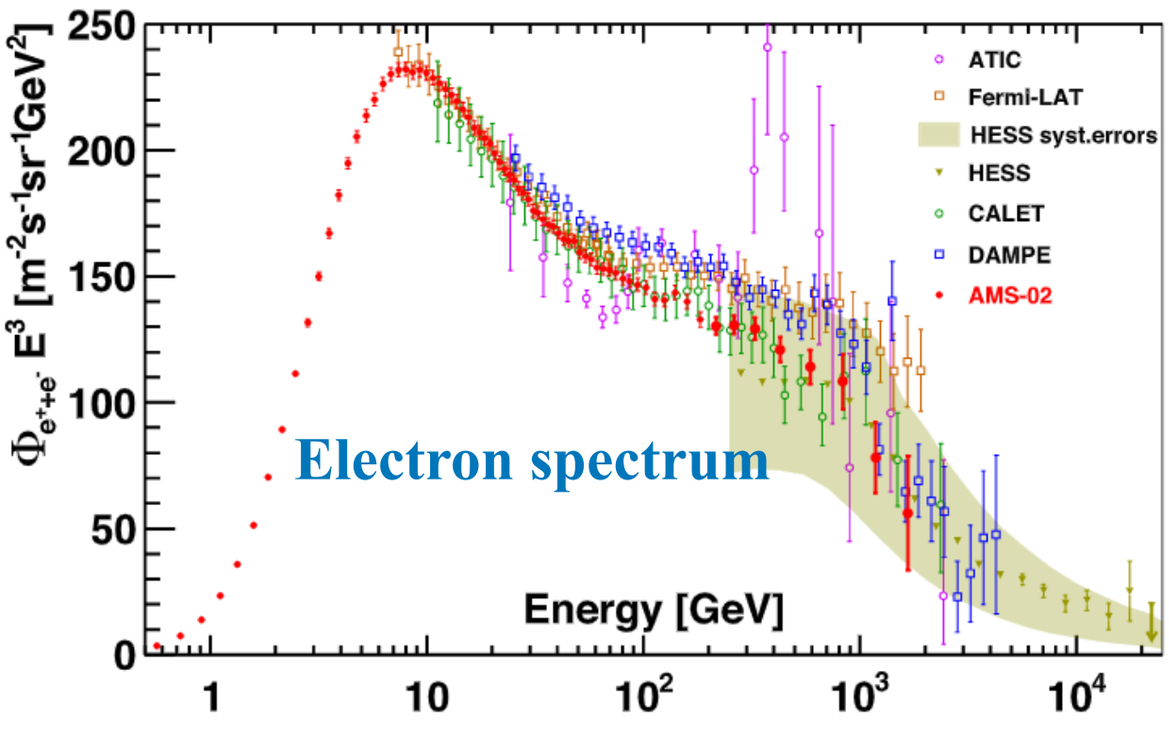
AMS-02 Collaboration, 2021, Physics Reports, 1

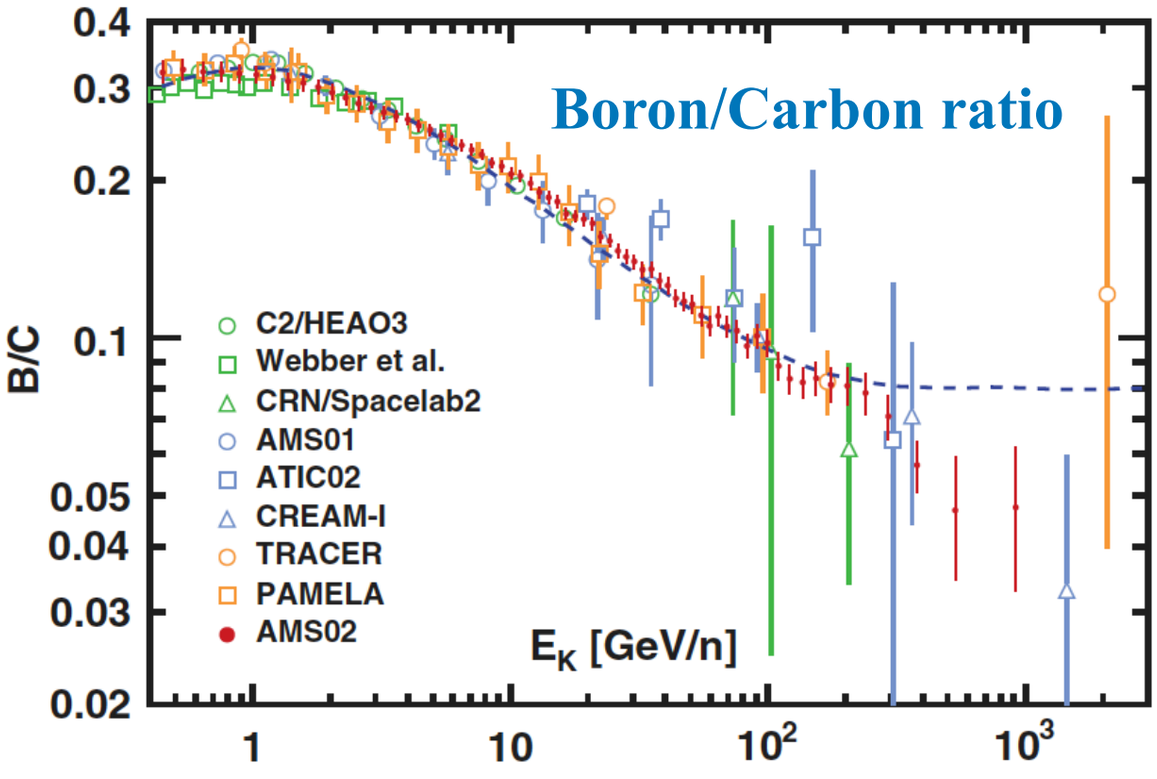
Measurements techniques: Spaced-based experiments

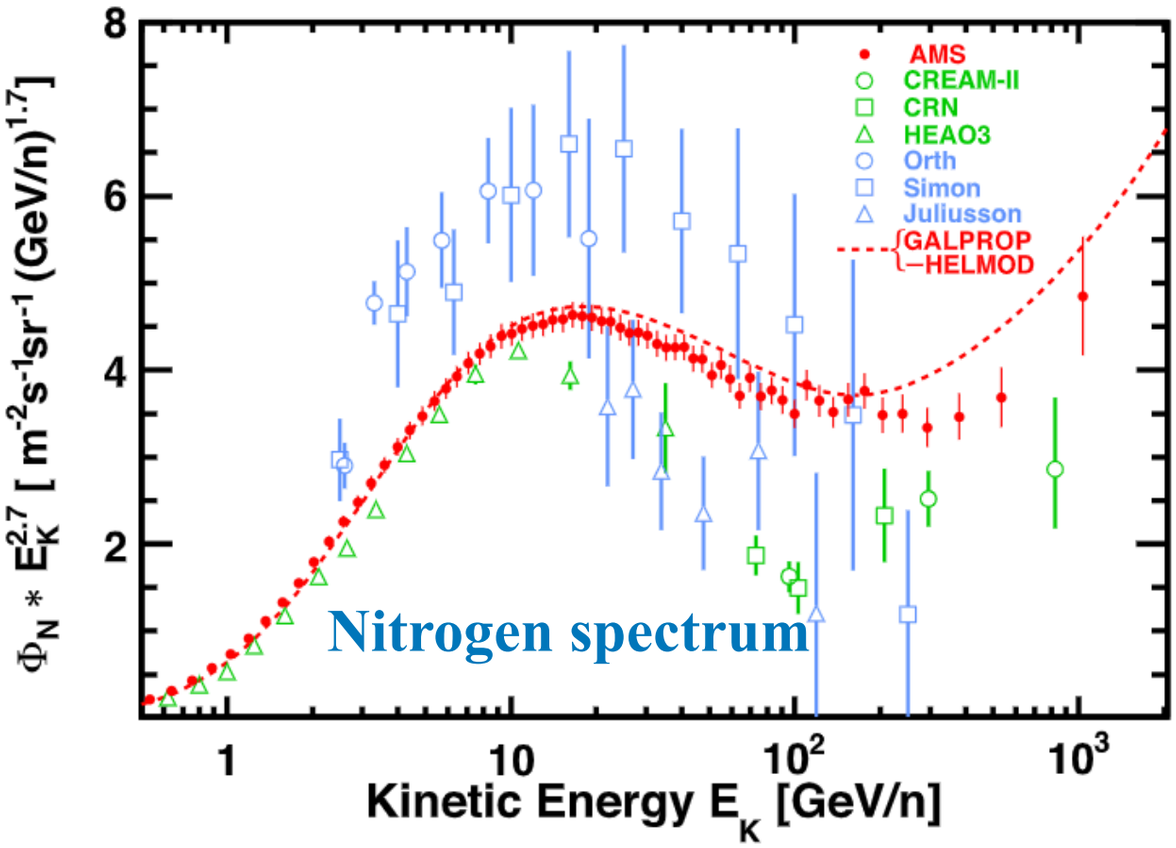
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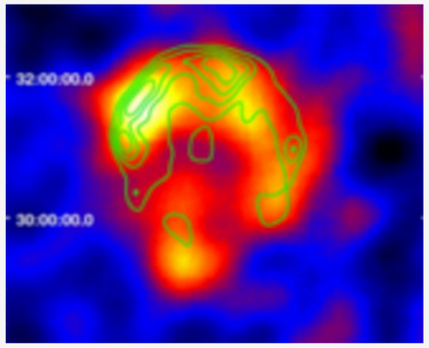




AMS-02 Collaboration, 2021, Physics Reports, 1

Measurements techniques: Spaced-based experiments

The *FERMI* Gamma-ray Space Telescope



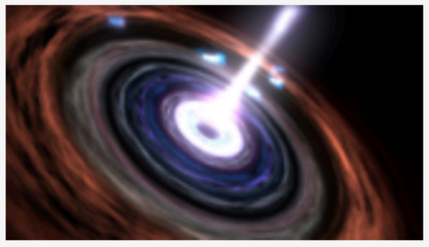
Supernova Remnants



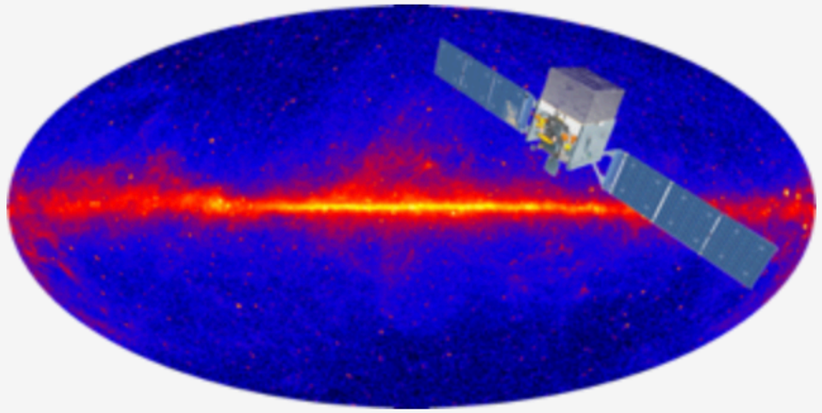
Gamma-ray Bursts



Pulsar Wind Nebulae

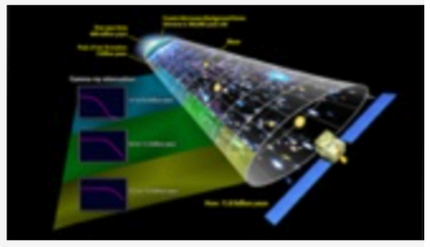


Active Galactic Nuclei

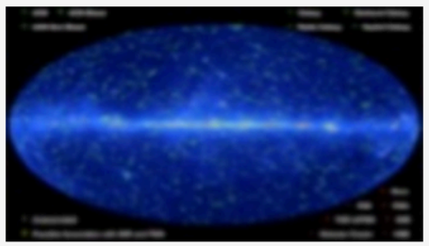


About Fermi

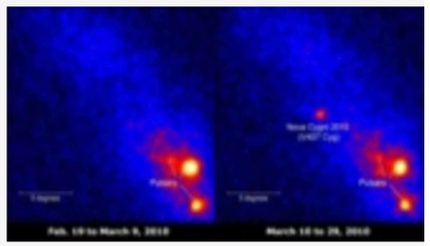
Click on the images or topic name for information about these science topics.



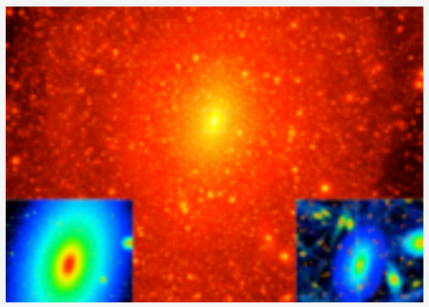
Extragalactic Background



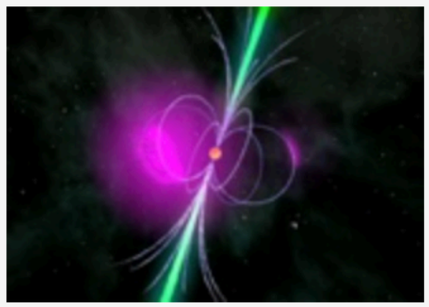
Catalogs



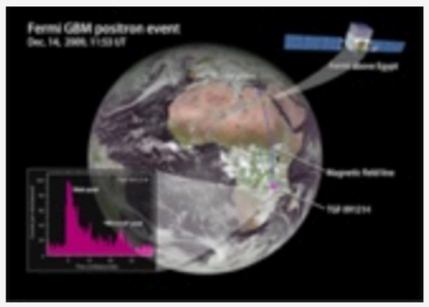
Binary Sources



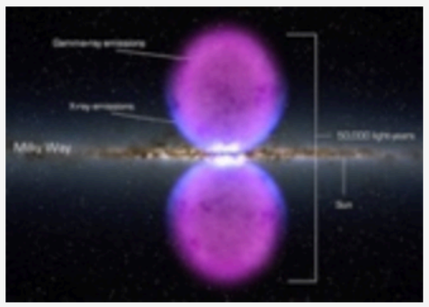
Dark Matter



Pulsars



Terrestrial Gamma-ray Flashes



Diffuse Gamma Radiation

https://fermi.gsfc.nasa.gov/science

Measurements techniques: Spaced-based experiments

FERMI telescope consists of two types of detectors:

Large Area Telescope (Primary instrument)
Pair-production instrument
Energy range: 20 MeV to ~300 GeV
Field of view: 2.4 steradians
Single photon angular resolution: <1° at 1 GeV
Timing accuracy: 1 microsecond

Gamma-ray Burst Monitor (Complementary instrument)
NaI and BGO scintillators
Energy range: 8 keV to 40 MeV
Field of view: 9.5 steradians
Gamma-ray burst localization: typical 3°
Timing accuracy: 2 microseconds

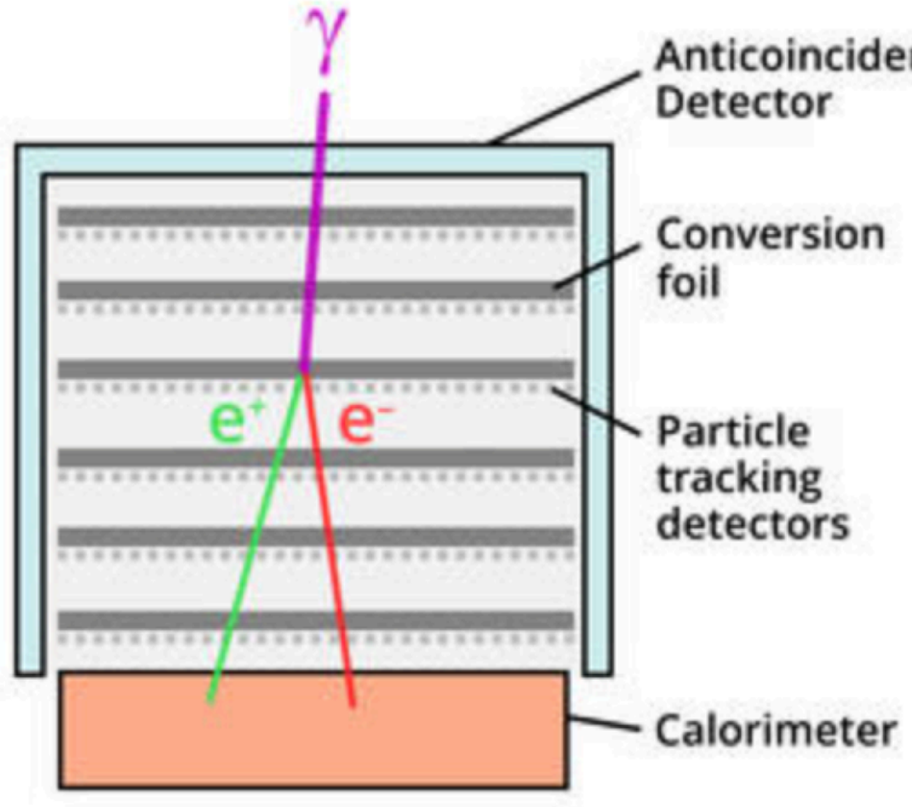


https://fermi.gsfc.nasa.gov/science

Measurements techniques: Spaced-based experiments

Components of Large Area Telescope (LAT) Components:

Anticoincidence Detector (ACD), used for CR background rejection:
The Anticoincidence Detector is the first line of defense against cosmic rays. It consists of specially formulated plastic tiles that produce flashes of light when hit by charged-particle cosmic rays (but not by gamma rays, which are electrically neutral). The ACD forms a "hat" that fits over the tracker.



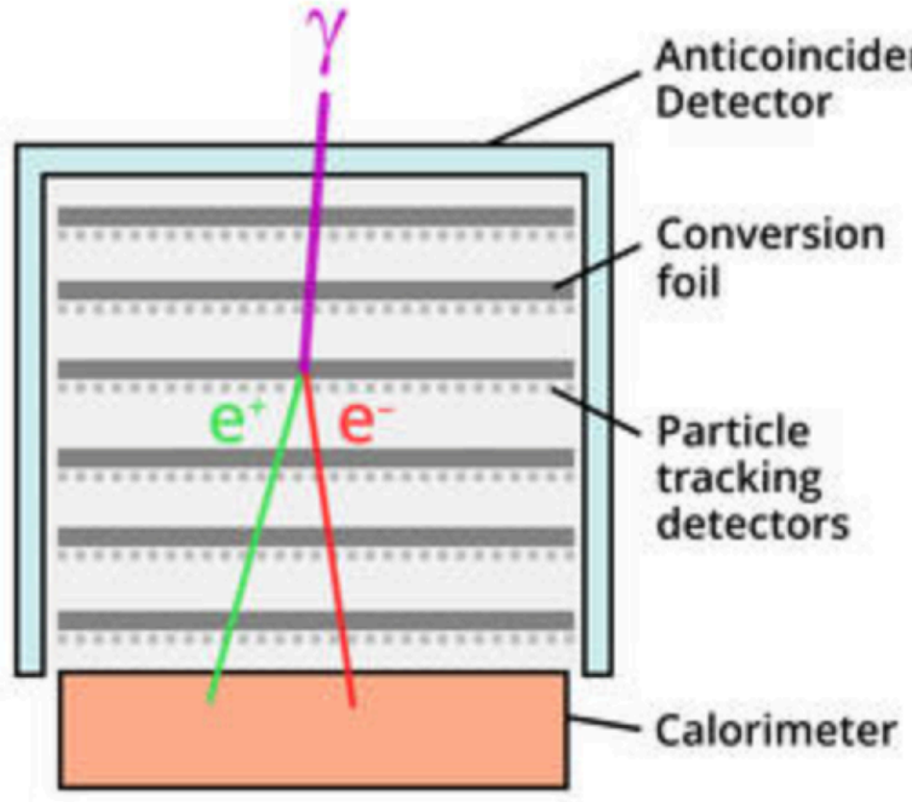
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Tracker, used to reconstruct direction of gamma rays:
The Tracker consists of a four-by-four array of tower modules. Each tower module consists of layers of silicon-strip particle tracking detectors interleaved with thin tungsten converter foils. The conversion foils transform an incoming gamma ray into a pair of electron and positron. The tracker tracks these particles as they travel through the instrument, enabling a reconstruction of the gamma ray's original direction. The pair-conversion signature is also used to help reject the much larger background of cosmic rays.



https://fermi.gsfc.nasa.gov/science

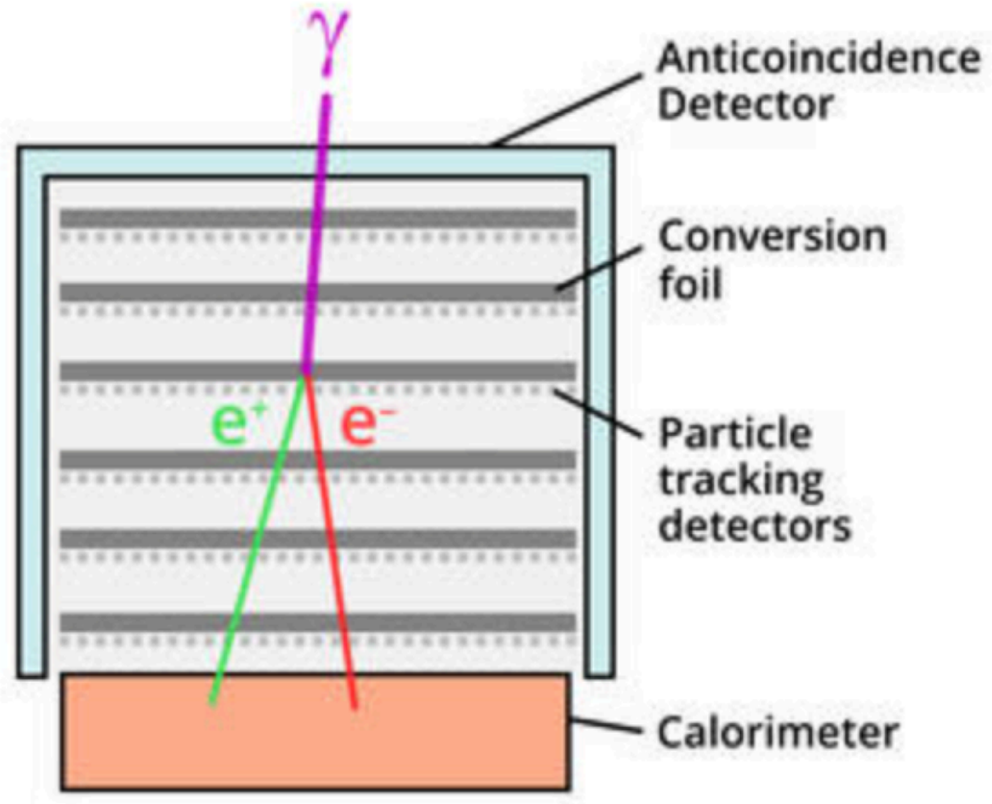
Measurements techniques: Spaced-based experiments

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Calorimeter, used to determine the gamma-ray energy:
The Calorimeter measures the energy of a particle when it is totally absorbed. The LAT Calorimeter is made of a material called cesium iodide that produces flashes of light whose intensity is proportional to the energies of the incoming particle. The Calorimeter also helps to reject cosmic rays, since their pattern of energy deposition is different from that of gamma rays.

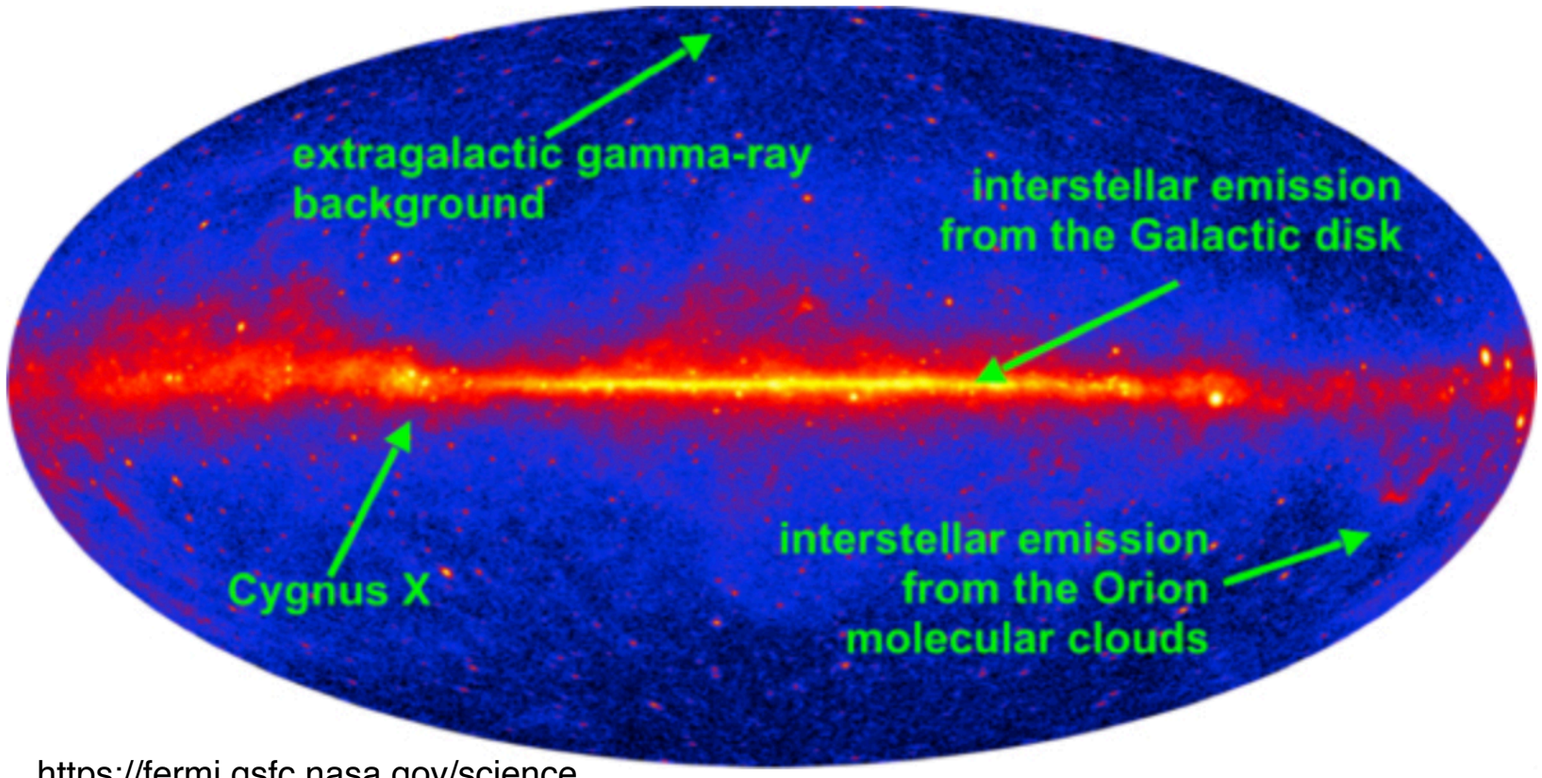


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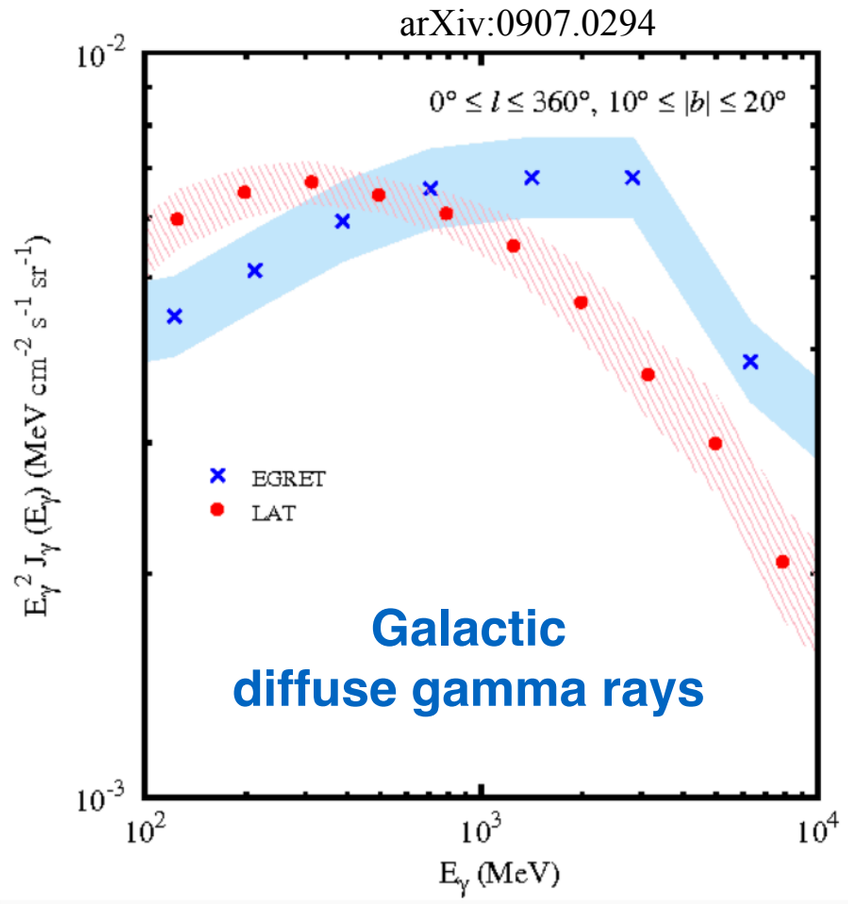
Measurements techniques: Spaced-based experiments

Some major science results from the *FERMI* experiment

Diffuse gamma rays



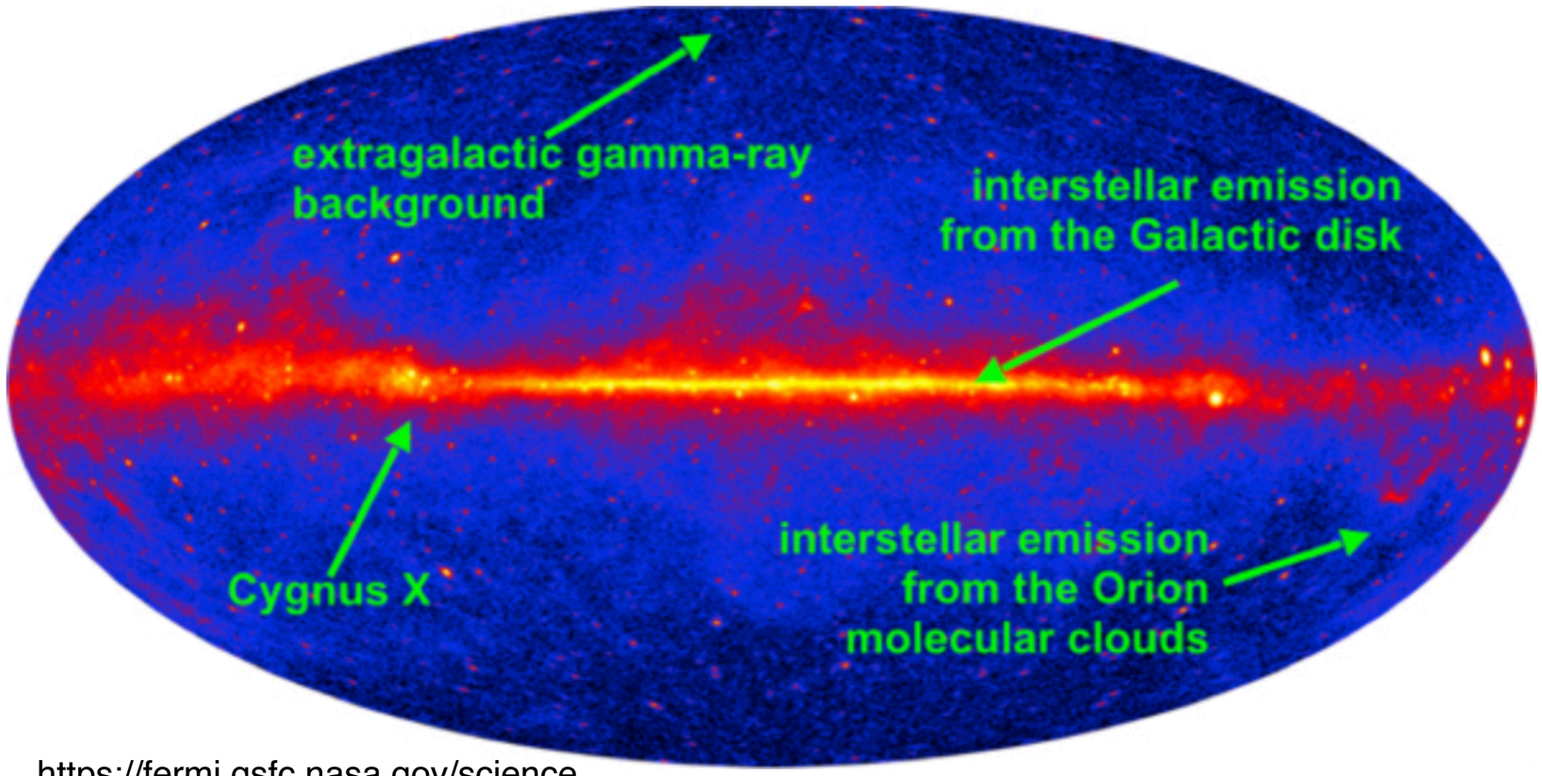
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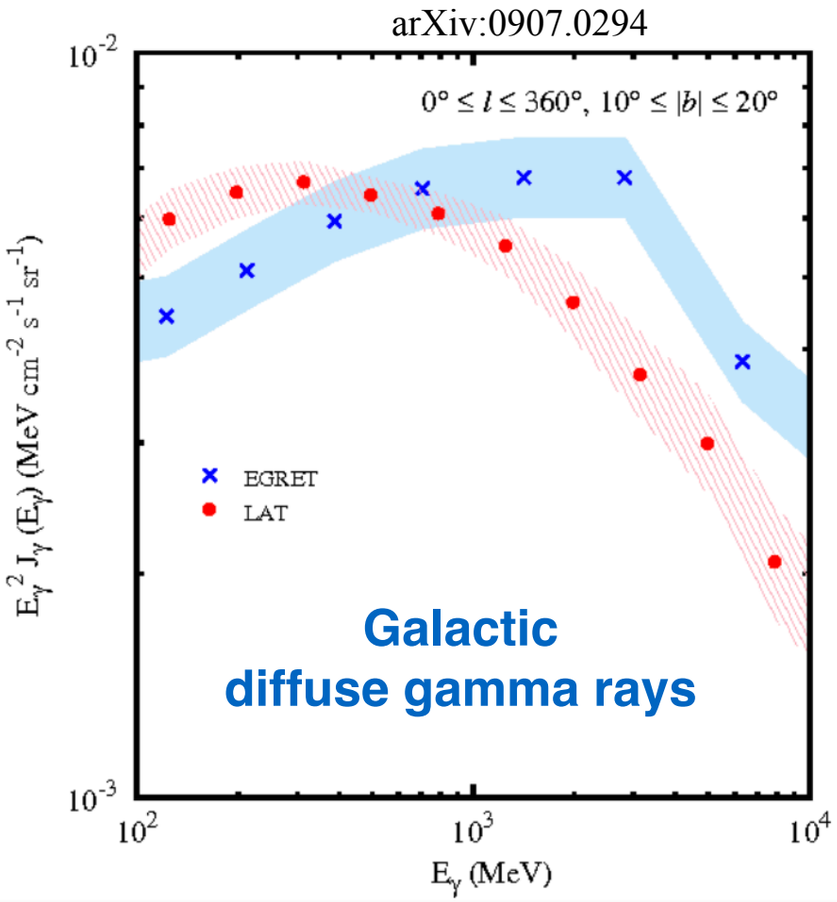
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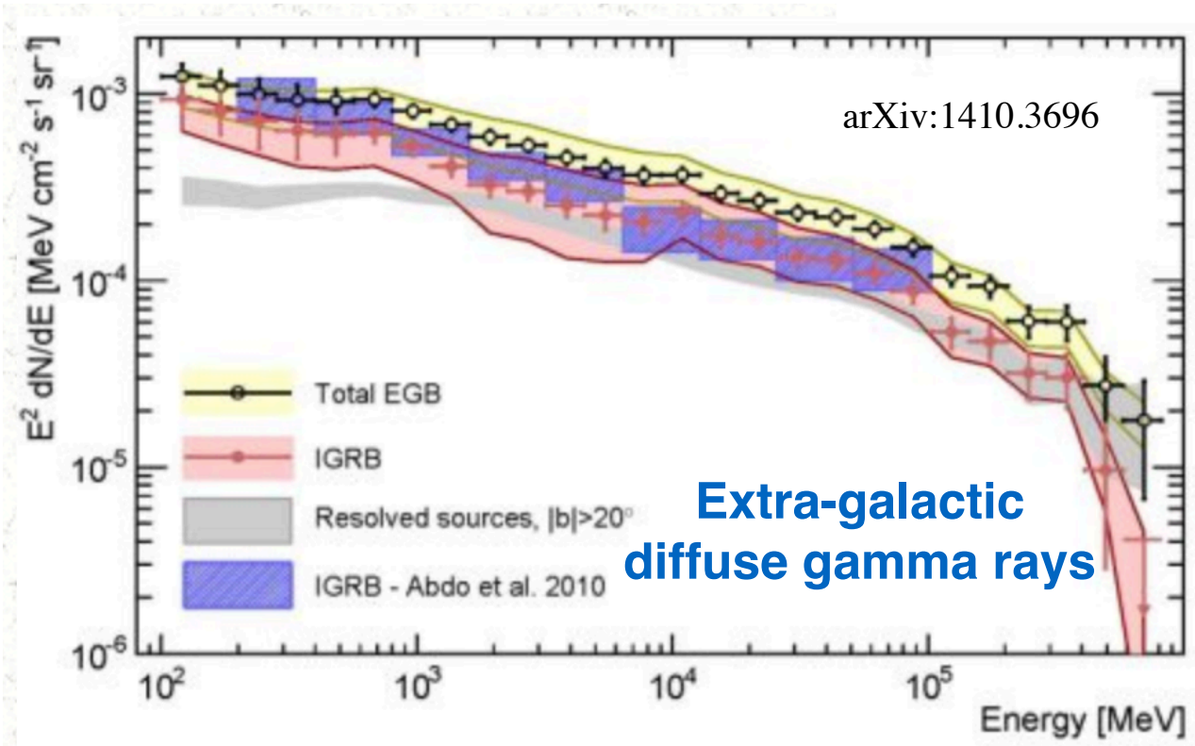
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Diffuse gamma rays



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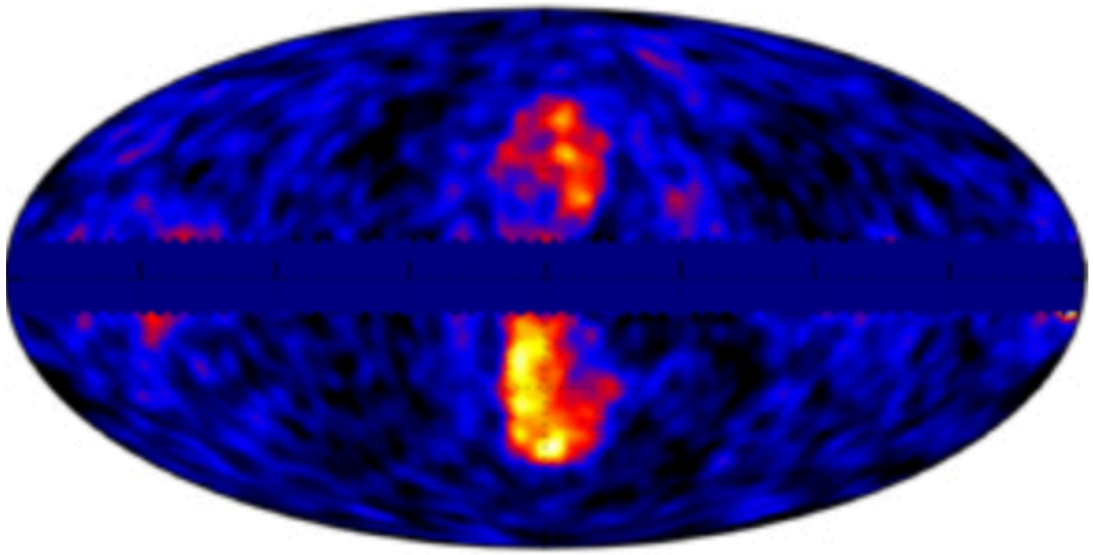


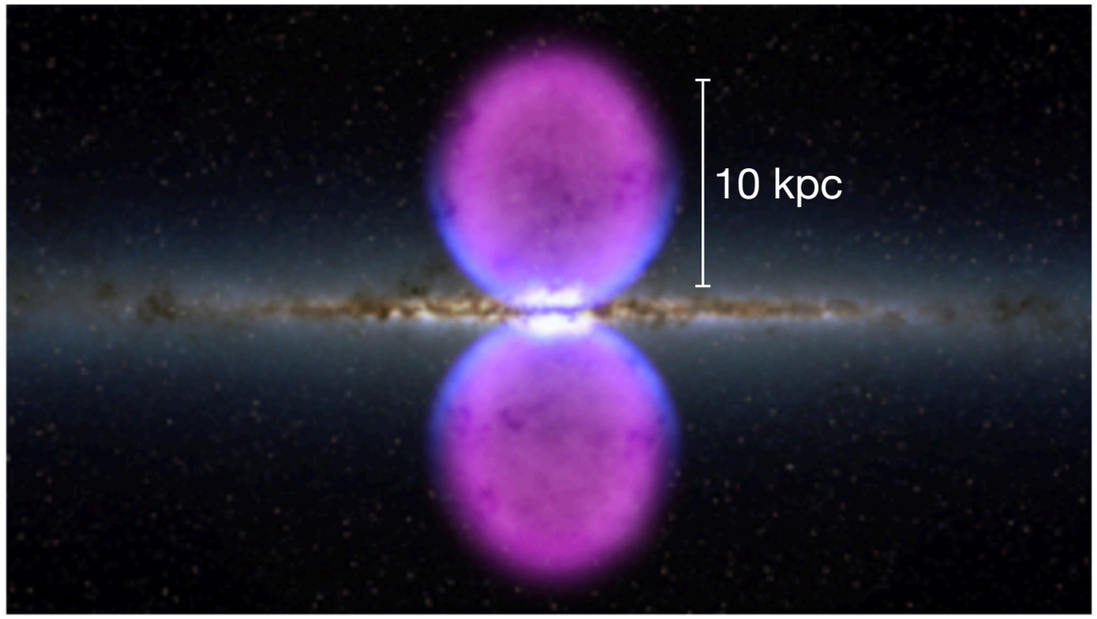
Measurements techniques: Spaced-based experiments

Some major science results from the *FERMI* experiment

Fermi Bubbles

https://fermi.gsfc.nasa.gov/science



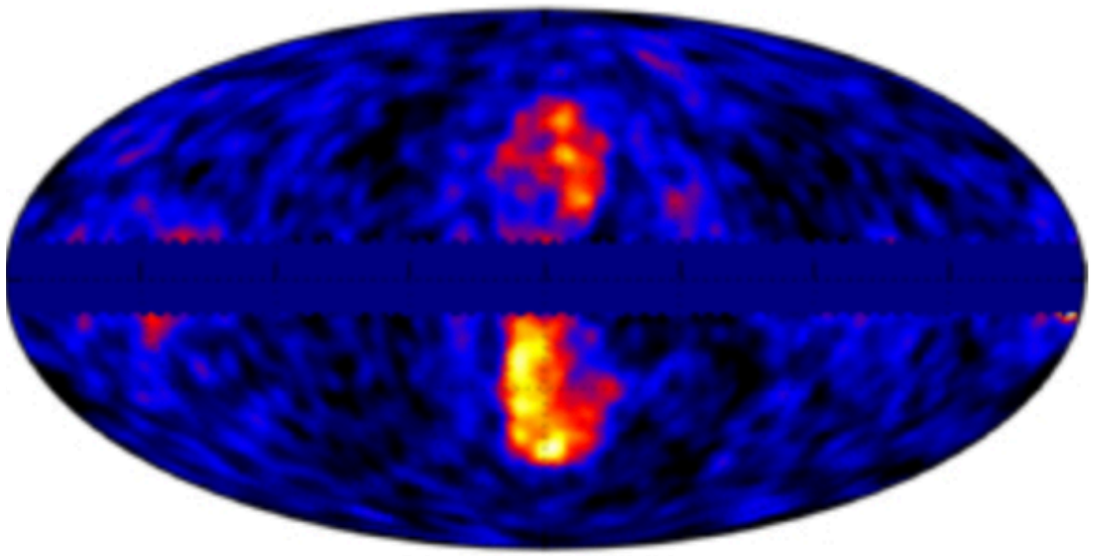


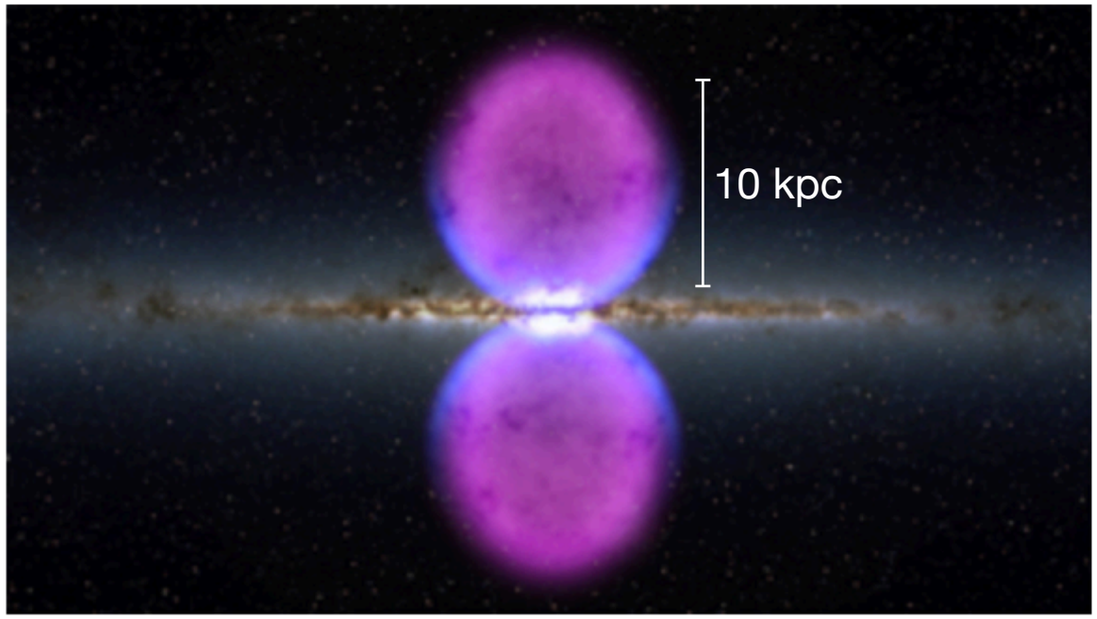
Measurements techniques: Spaced-based experiments

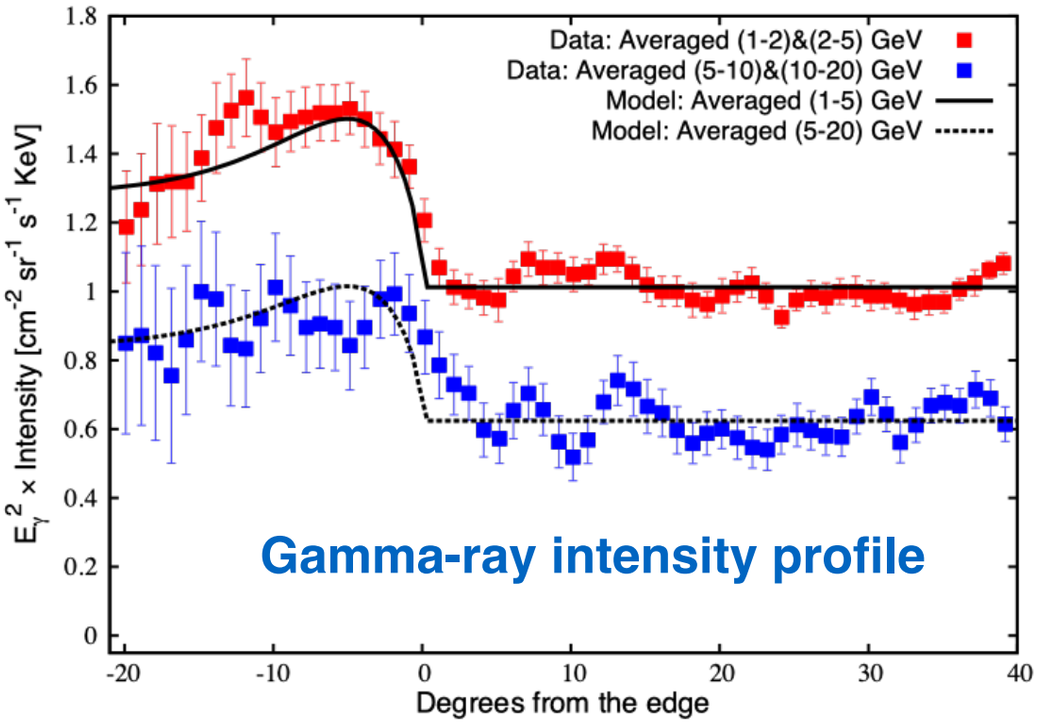
Some major science results from the *FERMI* experiment

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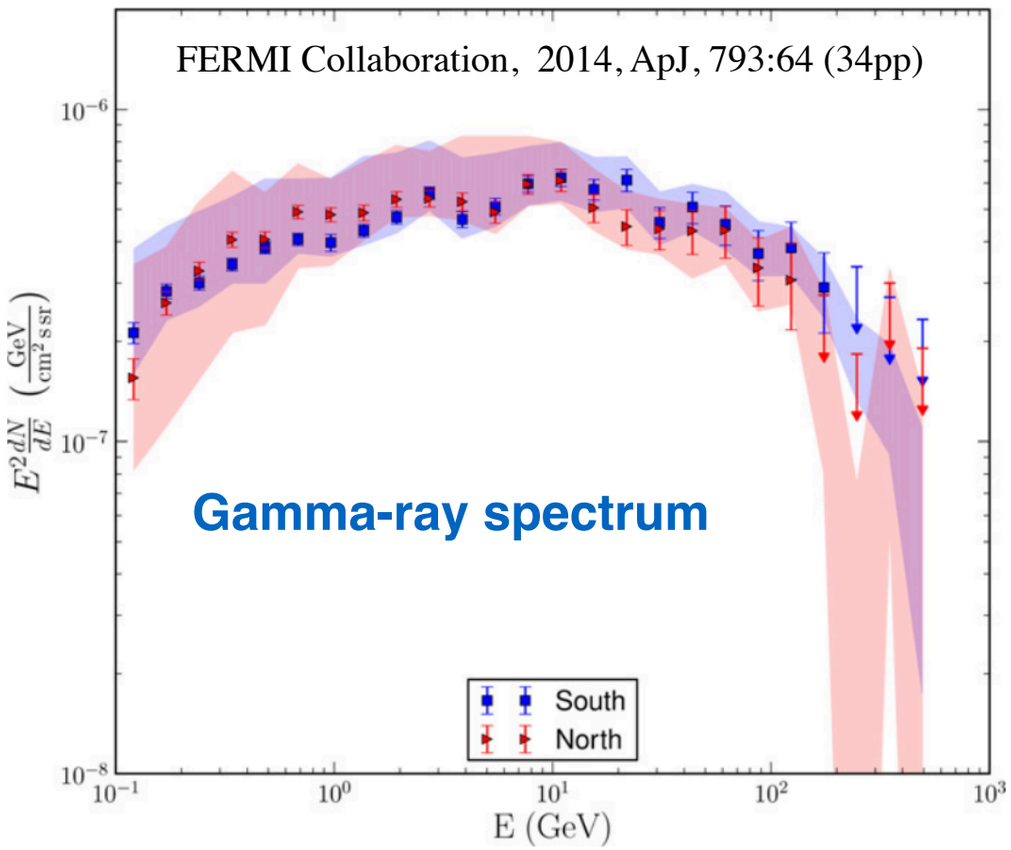
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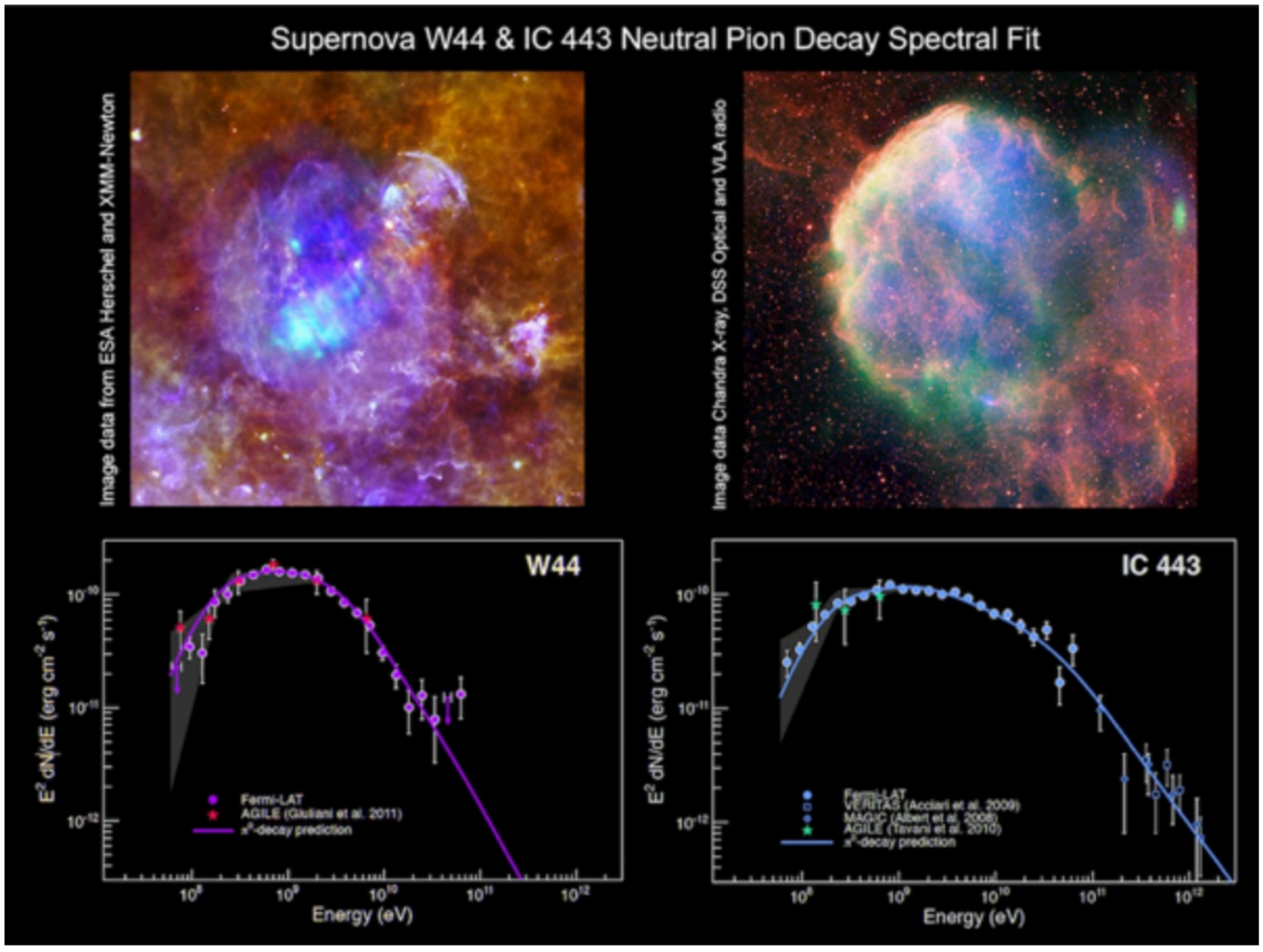
Thoudam 2013, ApJL, 778, L20



Measurements techniques: Spaced-based experiments

Some major science results from the *FERMI* experiment

First direct proof of hadronic acceleration inside supernova remnants?



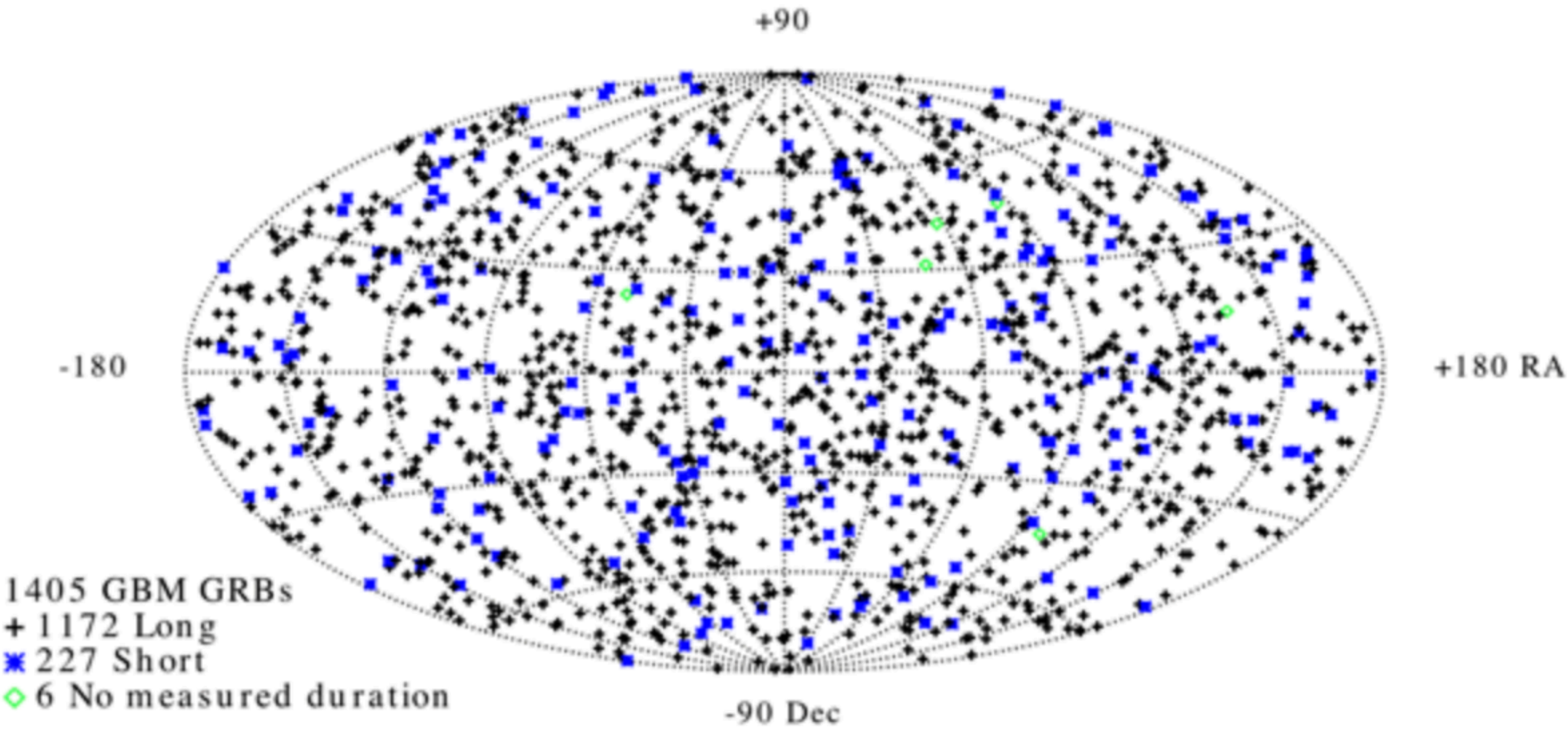
X-ray, optical, infrared and radio composite images (top row) of the interacting SNRs IC 443 and W44, along with their respective energy spectra measured by Fermi-LAT (bottom row). Both remnants show the distinctive steep rise in spectrum expected from pion-decay emission. (Credit: NASA/DOE/ Fermi LAT Collaboration, Chandra X-ray Observatory, ESA Herschel/XMM-Newton/DSS/VLA)

https://fermi.gsfc.nasa.gov/science

Measurements techniques: Spaced-based experiments

Some major science results from the *FERMI* experiment

Gamma-ray bursts measurements



Fermi GBM GRBs in the first six years of operation.

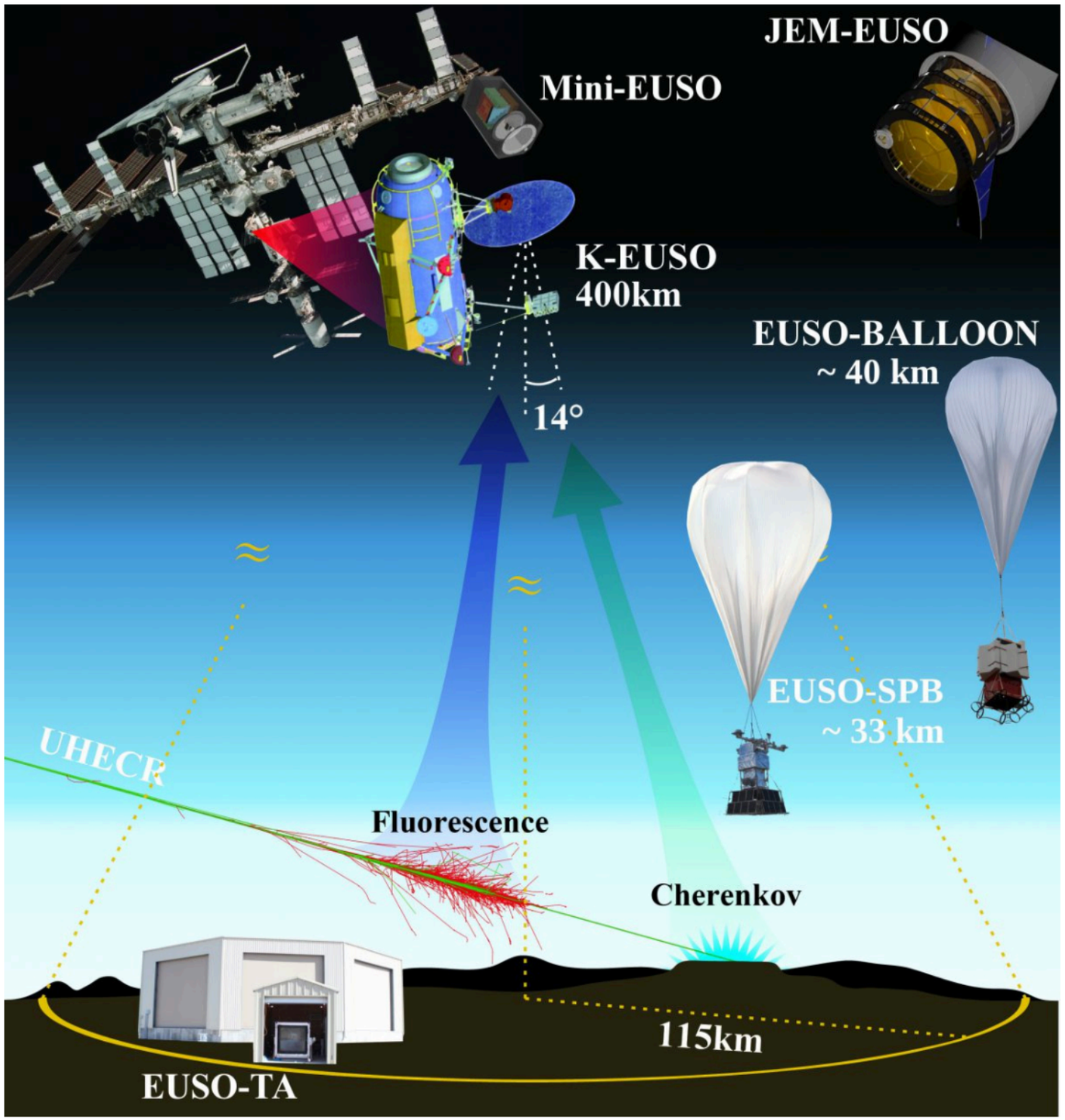
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Measurements techniques: Spaced-based experiments

The JEM-EUSO ultra-high-energy cosmic-ray detector

JEM-EUSO is proposed to be the first spaced-based detector for ultra-high energy cosmic rays above the GZK cut-off ($\sim 5.7\times10^{19}$ eV).

At these energies, ground-based experiments have limited performance due to the extremely low flux of particles (~ 1 particle/km2/sr/century).



https://www.jemeuso.org/missions/overview/

Measurements techniques: Spaced-based experiments

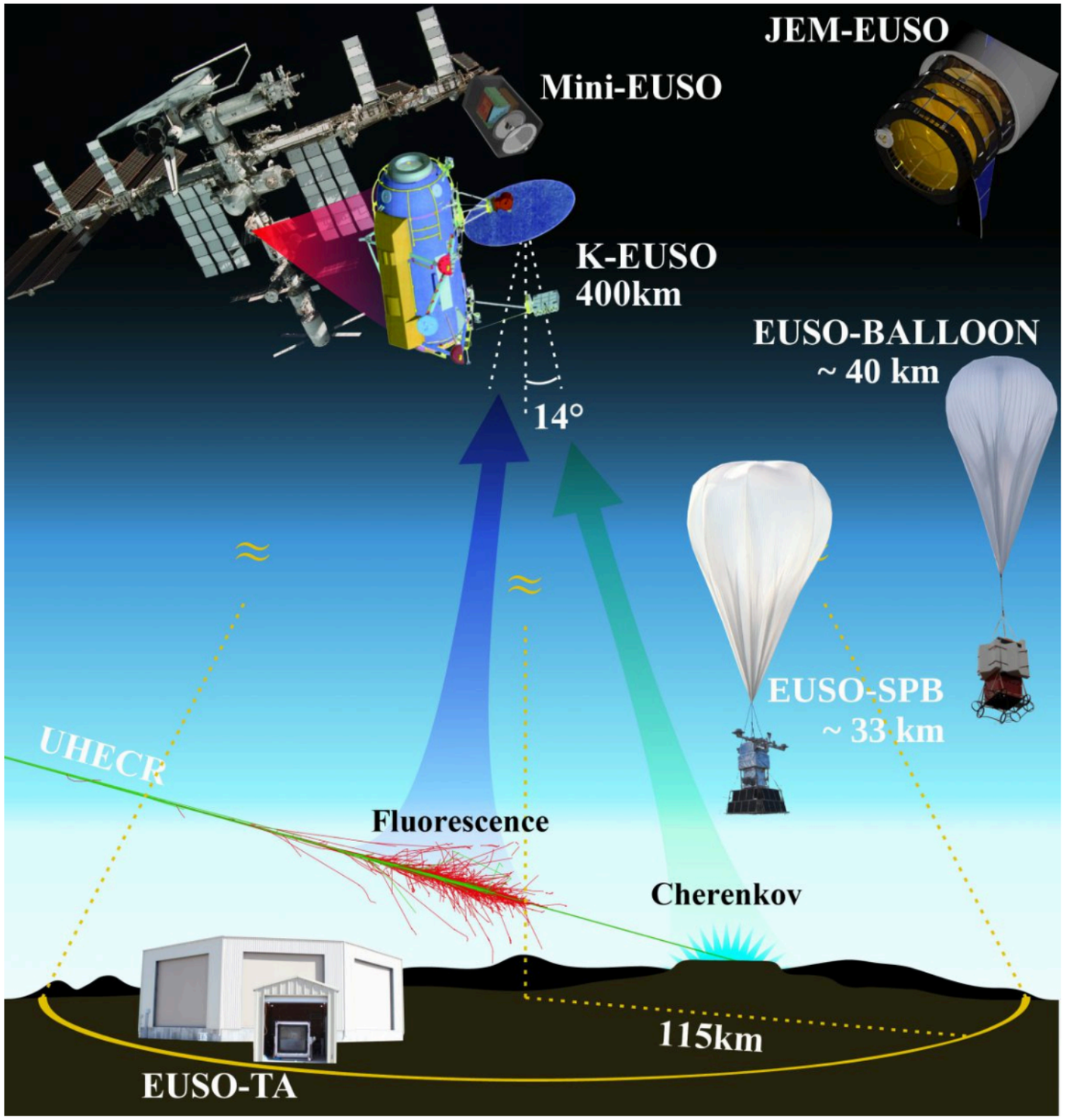
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JEM-EUSO is a two-ton refractor telescope using Fresnel lenses to detect the fluorescence light from cosmic-ray showers in the atmosphere.

The detector will provide complete and uniform coverage of the Celestial Sphere with an effective collection detector area that is several orders of magnitude larger than the largest ground-based detector.



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Measurements techniques: Spaced-based experiments

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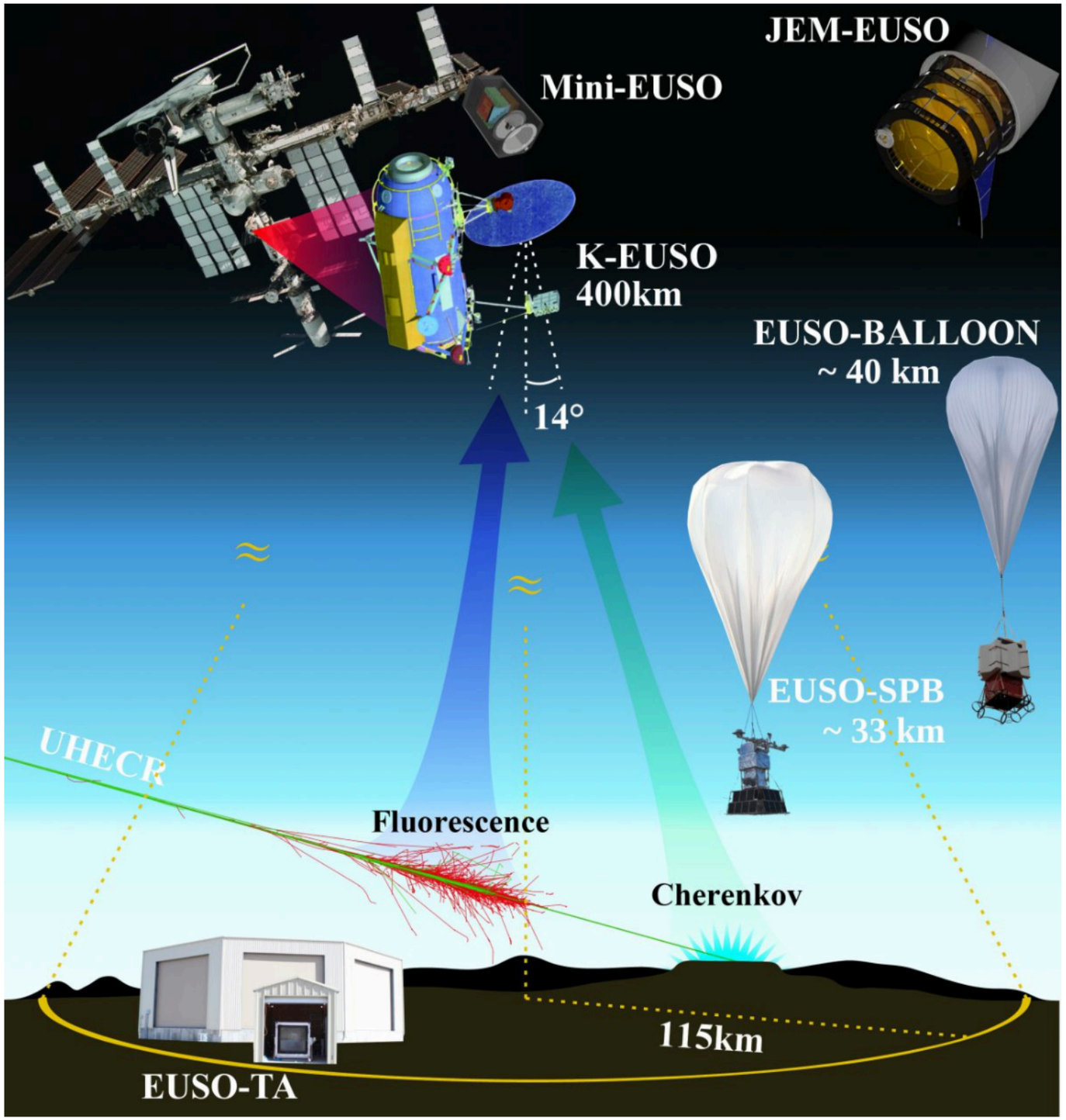
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Since 2013, a number of precursor projects on ground, balloon, and in space have been developed as shown in the figure.



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Next, we will learn about ground-based experiments.